VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI – 590018, KARNATAKA

Internship Report on

“<u>Study of 400 kV (GIS) / 220 kV (AIS) Substation and RTAMC</u>”

Submitted for the partial fulfilment of the requirements for the award of degree

BACHELOR OF ENGINEERING

In

ELECTRICAL AND ELECTRONICS ENGINEERING

Submitted by

J P JAIPUNEETH **USN: 1BY18EE025**

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In

“Power Grid Corporation of India Ltd”

Department of Electrical and Electronics Engineering

BMS Institute of Technology and Management, Bengaluru – 560 064

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INTERNSHIP CERTIFICATE

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

DEPT. OF ELECTRICAL AND ELECTRONICS ENGINEERING

CERTIFICATE

This is to certify that the internship report entitled "**Study of 400 kV (GIS) / 220 kV (AIS) Substation and RTAMC**" carried out by **J P JAIPUNEETH** bearing USN **1BY18EE025**, a bonafide student of BMS Institute of Technology and Management, in partial fulfilment for the award of Bachelor of Engineering in Electrical and Electronics of the Visvesvaraya Technological University, Belagavi during the year 2020-2021. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report deposited in the department library. The internship report has been approved as it satisfies the academic requirements in respect of Internship work prescribed for the said Degree.

College Guide	Dept. ICC Coordinator	Head of the Dept.
(Dr. Madhu Palati)	**(Dr. Nagaraj D C)**	**(Dr. N Ramarao)**

External Evaluators

1.

2.

ACKNOWLEDGEMENT

Training has an important role in exposing real-life situations in the industry.

To begin with, I wish to convey my heartfelt gratitude to **Almighty** for his help to complete the internship successfully.

I thank the management of Power Grid Corporation of India Ltd, Singanayakanahalli, for providing me this opportunity to complete my internship at their esteemed organization. I also thank **Mr. D R Murthy**, General Manager (HR), and **Mr. R. Shridharan**, Chief Manager (HR) for granting me permission to do my internship.

400 kV GIS /220 kV AIS Substation:

I thank **Mr. J P Jayaprakash**, Sr. General Manager (Station In-charge) for letting me visit, observe, and learn the 400/220kv GIS/AIS substation. I also thank Mr. Rahul, Deputy Manager for his guidance and support throughout the training period. I also thank **Mr. K. Shibin**, Junior Engineer, **Ms. Anitha Kumari Yadhav**, Junior Engineer, and staff for their valuable support in study of substation.

RTAMC (Regional Transmission Asset Management Centre):

I thank **Mr. G. Dhananjaya**, Sr. General Manager (RTAMC In-charge) for letting me visit, observe, and learn about RTAMC. I also thank **Mr. V. Murugan**, Sr. Deputy General Manager for his guidance and support throughout the training period. I also thank **Mr. M. Suresh Kumar**, Chief Manager, and other staff of RTAMC for their valuable support in study of RTAMC.

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Finally, I extend my sincere gratitude to my father, **Mr. Prabhu T**, Manager, POWERGRID for his continuous support, advice and guidance from initiation, till the completion of this Internship.

- J P JAIPUNEETH

Electrical and Electronics Department

BMS Institute of Technology & Management

VISION:

To emerge as one of the finest technical institutions of higher learning to develop engineering professionals who are technically competent, ethical and environment friendly for betterment of society.

MISSION:

Accomplish stimulating learning environment through high quality academic instruction, innovation and industry-institute interface.

Graduates of the program will,

PEO1	Have successful professional careers in Electrical Sciences, and Information Technology enabled areas and be able to pursue higher education.
PEO2	Demonstrate ability to work in multidisciplinary teams and engage in lifelong learning.
PEO3	Exhibit concern for environment and sustainable development.

After the successful completion of the course, the graduate will be able to,

PO1: **Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2: **Problem analysis**	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3: Design/development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4: **Conduct investigations of complex problems**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5: **Modern tool usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6: **The engineer and society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: **Environment and sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8: Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9: **Individual and team work**	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10: **Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11: **Project management and finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12: **Life-long learning**	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes:

The Graduates of the Program will be able to,

PSO1:	**Analyze and design electrical power systems.**
PSO2:	**Analyze and design electrical machines.**
PSO3:	**Analyze and design power electronic controllers for industrial drives.**
PSO4:	**Analyze and design analog and digital electronic systems.**

ABOUT COMPANY WHERE INTERNSHIP AVAILED

POWERGRID CORPORATION OF INDIA LTD

पावरग्रिड
POWERGRID

पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
POWER GRID CORPORATION OF INDIA LIMITED
(A Government of India Enterprise)

The Power Grid Corporation of India Limited was incorporated on 23 October 1989 under the Companies Act, 1956 with an authorized share capital of Rs. 5,000 Crore (subsequently enhanced to Rs. 10,000 Crore in Financial Year (FY) 2007–08) as a public limited company, wholly owned by the Government of India with 51.34% stake in the company as on 31 December 2020 and as principal electric power transmission company for the country.

Its original name was the "National Power Transmission Corporation Limited", was charged with planning, executing, owning, operating, and maintaining high-voltage transmission systems in the country. On 8 November 1990, the firm received its Certificate for Commencement of Business. Their name was subsequently changed to Power Grid Corporation of India Limited, on 23 October 1992.

- A "Maharatna" Central Public Sector Enterprise. **[1]**
- India's largest Electric Power Transmission Utility
- Listed Company since 2007: 51.34% holding by Government of India and balance 48.66% by public.
- Consistently rated "Excellent" under Memorandum of Understanding with Ministry of Power since 1993-94
- Operates ≍ 90% of Inter-State / Inter-Regional networks
- All-India inter-regional capacity ≍ 110,750 MW
- **International Ratings (consistent with Gol's Sovereign rating)**
    - Standard & Poor's - BBB- (Outlook-Stable)
    - Fitch - BBB- (Outlook-Stable)
    - Moody's – Baa2- (Outlook-Stable)
- **Domestic Ratings**
    - CRISIL - CRISIL AAA/Stable (Highest Safety)
    - ICRA - [ICRA] AAA/Stable (Highest Safety)
    - CARE - CARE AAA (Highest Safety/Lowest Credit Risk)

- **Transmission** (Key Statistics (As on August 31, 2021))
    - 169,397 CKm Transmission Lines
    - 261 Sub-Stations
    - >99 % System Availability
    - 445,860 MVA Transformation Capacity
- **Consultancy** (Key Statistics (As on August 31, 2021))
    - Transmission related consultancy to more than 150 domestic clients
    - Global footprints in 20 countries catering more than 25 clients.
- **Telecom** (Key Statistics (As on August 31, 2021))
    - Owns and operates ≍ 71,673 km of Telecom Network
    - Points of Presence in 458 locations & Points of Interconnections in 780 locations
    - Intra City network in 256 cities across India
    - Backbone Telecom Network Availability > 99.5%

[1] Taken from POWERGRID Official Website –

https://www.powergridindia.com/company-overview-0

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LIST OF FIGURES

1. SUBSTATION

- An electric power substation is a facility that provides a junction between parts of POWERGRID.
- It is the part of a power system, in which the voltage is transformed from high to low or low to high for transmission, distribution, transformation and switching.

1.1. FUNCTIONS OF A SUBSTATION:

- Protection of transmission system.
- Controlling the Exchange of Energy.
- Ensure steady State & Transient stability.
- Load shedding and prevention of loss of synchronism. Maintaining the system frequency within targeted limits.
- Voltage Control, i.e., reducing the reactive power flow by compensation of reactive power, tap-changing.
- Securing the supply by proving adequate line capacity.
- Data transmission via power line carrier for the purpose of network monitoring; control and protection.
- Fault analysis and pin-pointing the cause and subsequent improvement in that area of field.
- Determining the energy transfer through transmission lines.
- Reliable supply by feeding the network at various points.
- Establishment of economic load distribution and several associated functions.

1.2. CLASSIFICATION OF SUBSTATION:

- The substations can be classified in the following ways:
  - Based on Voltage Levels:
    - AC Voltage:
      - Extra Low voltage: Voltage level of below 70 Volts.
      - Low Voltage: Voltage level is between 70 Volts to 600 Volts
      - Medium Voltage: Voltage level between 0.6kV – 33 kV.
      - High Voltage: Voltage level between 33kV to 220 kV.
      - Extra High Voltage: Voltage level between 220kV to 760 kV.
      - Ultra-High voltage: Voltage level above 800kV.

- DC Voltage:
    - High Voltage Direct Current (HVDC): Uses DC for transmission of bulk power over long distances.

➢ Based on Location:

- Outdoor:
    - These substations are further subdivided into two categories
    - Pole Mounted Substations – Such Substations are erected for distributions of power in the localities. Single stout pole or H-pole and 4-pole structures with relevant platforms are operating for transformers of capacity up to 25 KVA, 125 KVA, and above 125KVA.
    - Foundation Mounted Substations – Such types of substations are used for mounting the transformers having capacity 33,000 volts or above
- Indoor:
    - In such type of substations, the apparatus is installed within the substation building. Such type of substations is usually for the voltage up to 11 KV but can be raised for the 33 KV or 66 KV when the surrounding air is polluted by dust, fumes, or gasses.

➢ Based on Configuration:

- Air Insulated Substation:
    - The main circuit potential is insulated from the ground by air using porcelain or composite insulators and/or bushings.
    - AIS is fully composed from air-insulated technology components.
    - AIS is the most common type of substation, accounting for more than 70% of substations all over the world.
- Gas Insulated Substation:
    - GIS is a high voltage substation in which the major conducting structures are contained within a sealed environment with a dielectric gas known as SF_6 gas (which is having highest properties of any dielectric media) as the insulating medium.
- Composite Substation:
    - Are substations having combination of the above two types of substations.

- Based on Application:
    - Step-up Type Substation
        - This type of substation gets the power supply from a near producing facility.
        - It uses a large power transformer for enhancing the voltage level for transmitting to the remote locations.
        - In this substation, the power transmission can be done by using a transmission bus to transmission lines.
        - This substation includes circuit breakers for switch generation as well as transmission circuits in & out of service as required.
    - Customer Substation
        - This type of substation works as the major source of power supply for one specific business client. The business case, as well as the requirements of technical, highly depends on necessities of customers.
    - System Stations:
        - This substation includes the huge amount of power transfer across the station.
        - These stations only offer no power transformers while others do voltage exchange as well. Typically, these stations supply the endpoints to the transmission lines creating from switchyards & supply the electrical energy for circuits that supply transformer stations.
        - They are important to long-term consistency. These stations are strategic services as well as very costly to build as well as to maintain.
    - Distribution Type Substation:
        - Distribution type substations are placed where the main voltage distributions are stepped down to supply voltages to the consumers using a distribution network. The voltage of any two phases will be 400 volts, and the voltage between neutral and any phase will be 230volts.

- Step-down Type Substation:
    - They can connect different parts of the network and that are a source of sub-transmission or distribution lines.
    - This type of substation can change the transmission voltage to a sub-transmission voltage (Ex. 66kV). The converted voltage lines can provide a source for distribution substations.
    - In some cases, power is tapped from the line of transmission line to utilize in an industrial capacity along the way.
- Underground Distribution Substation:
    - Installing the substation underground decreases requirement of space and the surface area, which can also be used for other constructions like buildings, shopping malls, etc.
    - The main concept of the underground substation is to offer the best conventional substation by reducing the space occupied above land.
- Switchyard:
    - The switchyard is the mediator among the transmission as well as generation, and equal voltage can be maintained in the switchyard.
    - The main purpose of this is to supply the generated energy from the power plant at the level of voltage to the nearby transmission line or National Grid (POWERGRID).

1.3. REQUIREMENTS OF A SUBSTATION:

❖ **Factors for Selection of Site:**

➢ Type of substation:

- The category of the substation is important for its location. For example, a step-up transformer is a point where power from various sources is pooled and step up for long distance transmission should be located as close as possible to minimise the losses. Similarly, the step-down transformer should be located nearer to the load centre to reduce transmission losses, the cost of the distribution system and better reliability of supply.

- Availability of suitable and sufficient land:
  - The land selected for a substation should be level and open from all sides. It should not be waterlogged particularly in the rainy season. The site selected for substation should be such that approach of transmission lines and their take off can be easily possible without any obstruction. The places nearer to airdrome, shooting practice ground, etc. should be avoided.
- Communication facility:
  - Suitable communication facility is desirable at a proposed station, both during and after its construction. It is better, therefore, to select the site alongside the existing road to facilitate an easier and cheaper transportation.
- Atmospheric Pollution:
  - The atmosphere around the ground factories produces metal corroding gas, air fumes, conductive dust, etc. And the area near the sea coast may be more humid and is harmful to the proper running of the power system. Thus, the substation should not be located near the factories or sea coast.
- Availability of Essential Facilities to the Staff:
  - The site should be such where staff can be provided essential facilities like school, hospital, drinking water, housing, etc.
- Drainage Facility:
  - The site selected for the proposed substations should have proper drainage arrangement or the possibility of making effective drainage, avoid pollution of air and growth of micro-organism and health.

Factors for Design of Substation:

- Building a new substation or retrofitting the old one is a complex process full of design and engineering tasks to be worked on.
- The main steps in substation design and engineering are as follows:
- Switching System:
  - Selection of a substation switching system: ring bus, breaker-and-a-half, etc. based on reliability requirements.
- Key Plan, Locations of Components:
  - Preparation of a key plan which should show the location of all components of a substation and their interconnections, as well as steel structures, control house, fire walls, driveways, fence and property line.

- Equipment Selection and Ordering:
    - Selection and ordering of equipment, is usually done in a utility company by a designated group of equipment experts.
    - These experts specify required ratings of transformer, breaker, etc.; request bids from approved vendors, evaluate the bids, place the order with a winning bidder, and participate in the testing and commissioning of the equipment.
- Engineering Support:
    - Engineering support for licensing and permitting which includes preparation of necessary drawings sealed by professional engineers, testifying at public hearings at the municipalities where a new substation is planned to be built, ordering of noise studies and selecting means of noise mitigation if needed.
- Civil and Structural Design:
    - Pile design
    - Foundations
    - Steel structures
    - Control house
- Electrical Layout Design:
    - Positioning of equipment.
    - Bus design.
    - Design of manhole and conduit system.
    - Design of auxiliary AC power system.
    - Selection of D.C. batteries and battery chargers.
    - Layout of control house.
    - Grounding and lightning protection design.
- Relay Protection, SCADA:
    - Relay protection & instrumentation system schematic & wiring diagrams.
    - Relay racks or panels.
    - Remote control & metering (SCADA – system control and data acquisition)
- Construction Support:
    - It includes a resolution of technical problems discovered during construction, ordering of additional materials, etc.

❖ **Equipment and Parts Required for the Substation:**

- Each sub-station has the following parts and equipment:
- Outdoor Switchyard:
  - Incoming Lines
  - Outgoing Lines
  - Busbar
  - Transformers
  - Bus post insulator & string insulators
  - Substation Equipment such as circuit-breakers, isolators, earthing switches, surge/lightning arresters, CTs, VTs, CVTs, neutral grounding equipment.
  - Station Earthing system comprising ground mat, risers, auxiliary mat, earthing strips, earthing spikes & earth electrodes.
  - Overhead earth wire shielding against lightning strokes.
  - Galvanised steel structures for towers, gantries, equipment supports.
  - PLCC equipment including line trap, tuning unit, coupling capacitor, etc.
  - Power cables.
  - Control cables for protection and control
  - Roads, Railway track, cable trenches
  - Station illumination system.
- Main Office Building:
  - Administrative building
  - Conference room etc.
- Switchgear and Control Panel Building:
  - Low voltage AC Switchgear
  - Control Panels, Protection Panels
- Battery Room and DC Distribution system:
  - D.C. Battery system and charging equipment
  - D.C. distribution system
- Mechanical, Electrical and other auxiliaries:
  - Firefighting system
  - Diesel Generator (D.G.) Set
  - Oil purification system

❖ Selection Details of Major Equipment:

- Selection of major equipment is the most critical tasks in substation engineering.
- Power Transformer Ratings:
  - Capacity including overload capability
  - Cooling class
  - Frequency
  - Primary and secondary voltage
  - Phase relation between primary and secondary voltages
  - BIL for both high and low voltage sides
  - Voltage regulation requirements: load and no-load taps
  - Transformer impedance
  - Sound level
- Circuit Breaker Rating:
  - Rated maximum voltage
  - Rated continuous current
  - BIL (Basic Insulation Level)
  - Rated short circuit current
  - Interrupting time
  - Rated frequency
- Current transformer (CT) ratings:
  - BIL (Basic Insulation Level)
  - Rated current
  - Rated frequency
  - Number of taps and ratio for each tap
  - Accuracy class
  - Type (bushing CT, free standing, etc.)
- Voltage transformer (VT) or Potential transformer (PT) ratings:
  - Rated voltage factor
  - Rated primary voltage (Up)
  - Rated secondary voltage
  - Accuracy power
  - Accuracy class
  - Voltage ratio error
  - Phase or phase displacement error
  - Rated thermal limiting output

➢ <u>Disconnect switch ratings</u>:

- Rated voltage
- Rated frequency
- Rated current
- Rated short-time withstand current and duration
- Rated peak-withstand current
- Rated short-time power frequency withstand voltage (rms) – To earth between breaks
- Rated lightning impulse withstand voltage (peak) – To earth between breaks
- Rated busbar transferring current
- Type of motor operation mechanism
- Switching capacitive current
- Switching inductive current
- Switching busbar transferring current Bus-transfer current

❖ **Single Line Diagram (SLD):**

➢ The single line diagram (SLD) is the most basic of the set of diagrams that are used to document the electrical functionality of the substation. Its emphasis is on communicating the functions of the power equipment and the associated protection and control system.

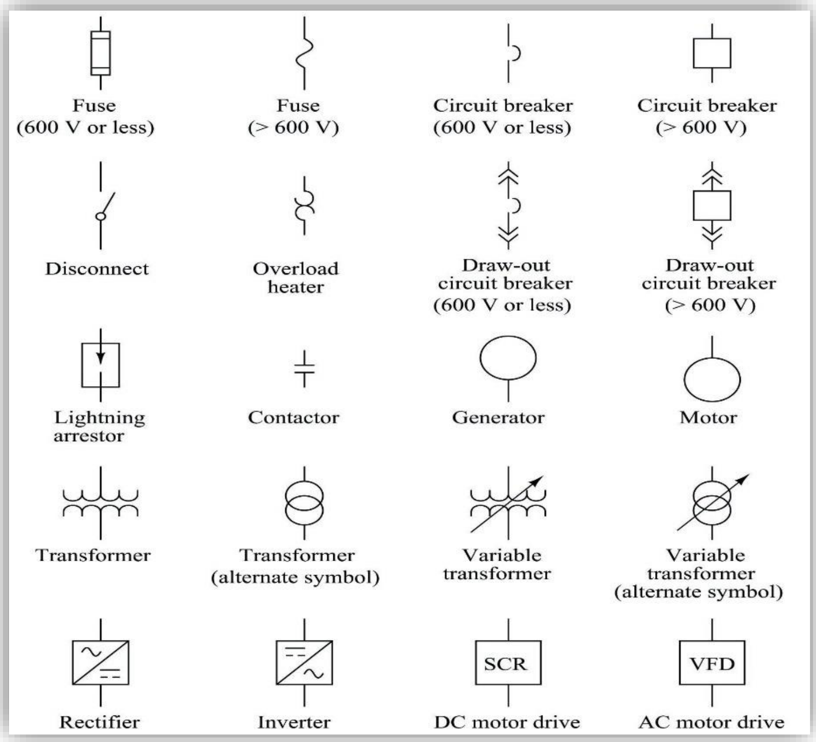


Fig 1. IEC standard symbol for equipment representation in SLD

Rules and Methodologies for a Substation:

- Always try to show a single line of diagram for all main electrical connection where Main Circuit Breaker or Vacuum Circuit Breaker (V.C.B), Current Transformer (C.T), Potential Transformer (P.T), Isolators, Earth Break Switch (A.B.S), Air Circuit Breaker (A.C.B), Busbar Trunking System (B.B.T) and Power Transformer symbols will display.
- The main layout for the switchgear keeping in view total capacity of the substation, all method of control, the different number of feeders, safety, reliability, simplicity, flexibility and the total amount of cost for making the substation will be displayed.
- All individual main circuits must be designed where a minimum amount of risk involved in the sudden failure.
- If there is any sudden failure then the main substation layout should be designed that can isolate any section.
- Access way for maintenance and inspection must be easier where all equipment can be inspected.
- Always maintain a partition or barrier between two units for avoiding serious types of troubles from spreading.
- To limit the short circuit current, a reactor can be used. It will rupture the capacity for a circuit breaker to become adequate.
- The circuit breaker range for any kind of circuit should be 2000A for avoiding using the large capacity of a circuit breaker.
- All arrangements must be made for fire extinguishing.
- The amount of earth conductor should be a sufficient cross-sectional area that really carries the fault current in-service time.
- Very effective and a proper automatic electrical protective gear must be used.
- All types of power cables must be separated for any kind of control cables.
- A fireproof switch room and a cable room should be used to avoid the fire hazard.
- An adequate arrangement must be made for handling the oil.
- Procedure for any kind of emergency action like an electrical shock should be displayed in a substation.

1.4. POWERGRID CORPORATION OF INDIA LTD. SUBSTATION YELAHANKA:

❖ About the Substation:

- POWERGRID Yelahanka is a 34 acres 400kV/220kV substation project, located at RTO road, Singanayakanahalli, Yelahanka, Bengaluru.
- The substation is designed for 400kV (GIS)/220kV (AIS) (Step-Down operation).

Fig 2. PowerGrid Substation

- The 400kV substation is designed as Gas Insulated Substation.

Fig 3. GIS 400kV Substation (Siemens)

- The 200kV substation is designed as Air Insulated Substation.

Fig 4. AIS 220kV

- The 400kV (GIS) substation has incoming lines from 4 places, i.e., Nelamangala, Devanahalli, Tumkur-I and Tumkur-II.

- The 220kV (AIS) substation has 10 outgoing lines, out of which only 2 are functioning at present, i.e., KPTCL, Puttenahalli circuit I & II.

- The incoming feeders are Overhead Transmission Lines.

Fig 5.1. Transmission Lines

- Whereas, the outgoing feeders are 220 kV, 2000 A capacity Underground Cables.

Fig 5.2. Under Ground Cables

❖ <u>Single Line Diagram of the AIS 400kV substation is given below</u>:

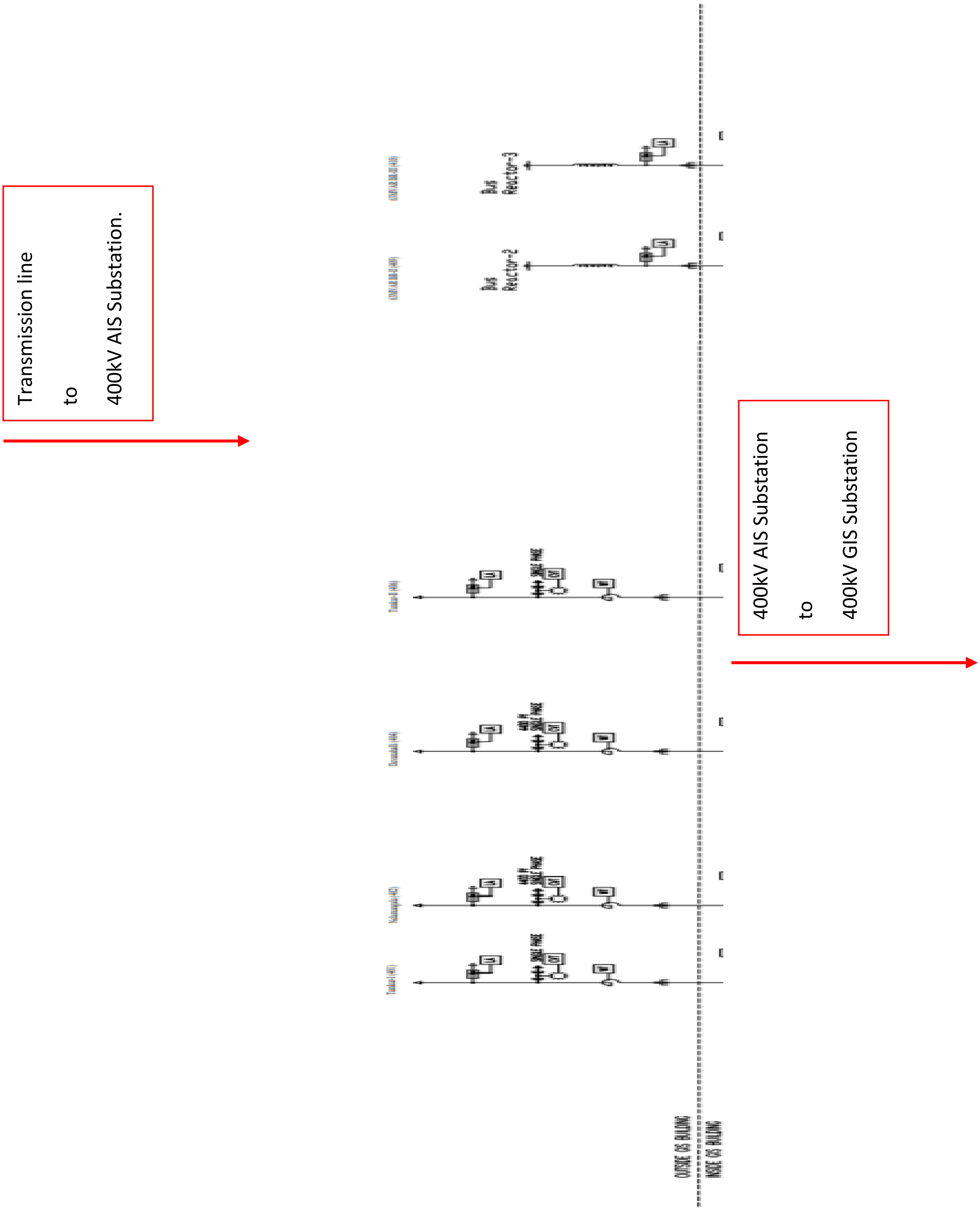


Fig 6.1. SLD of AIS 400kV

➢ <u>TAP HERE TO SEE THE COMPLETE SLD</u> .

❖ Single Line Diagram of the GIS 400kV substation is given below:

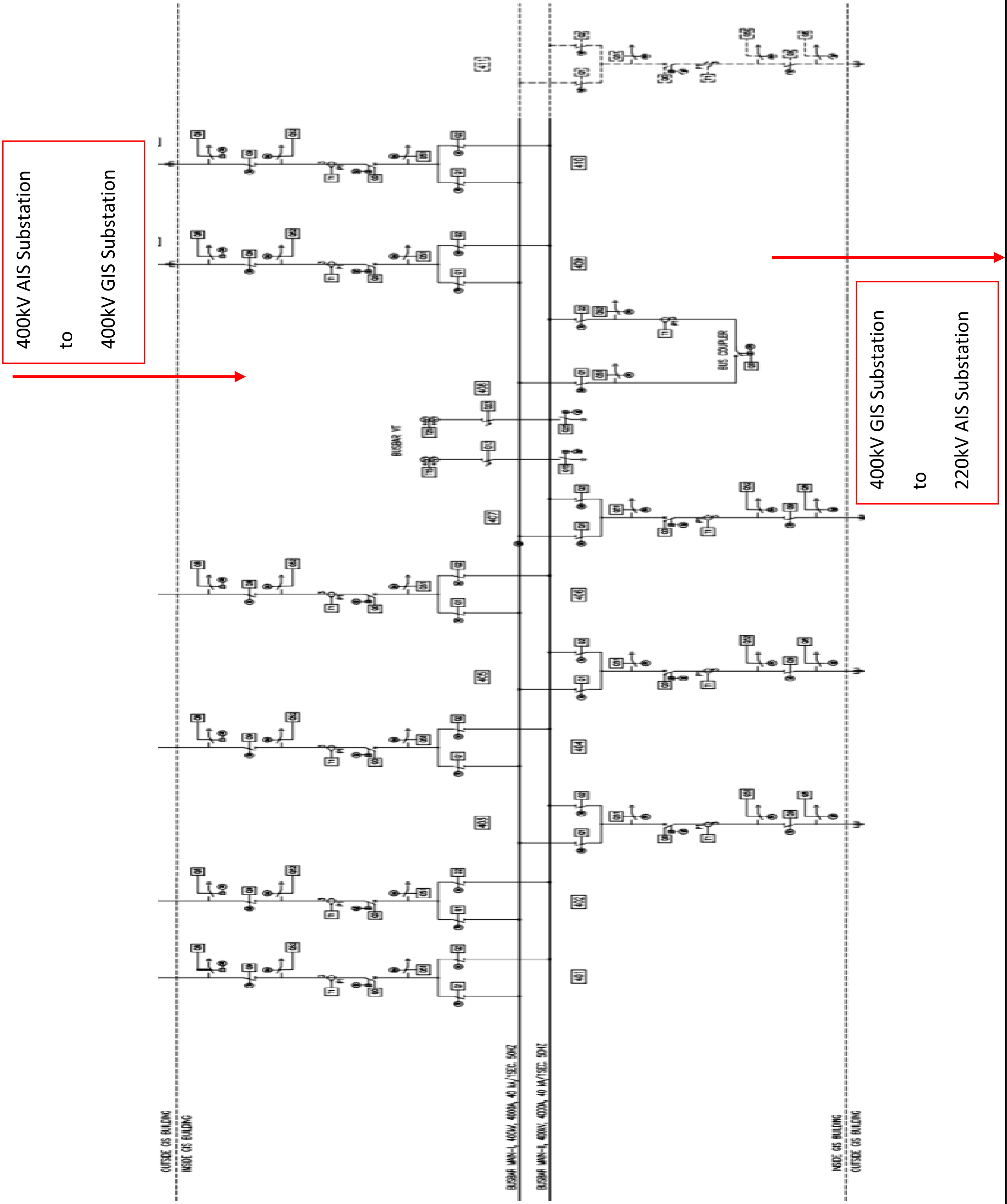


Fig 6.2. SLD of GIS 400kV

➢ TAP HERE TO SEE THE COMPLETE SLD .

❖ <u>Single Line Diagram of the 220kV AIS Substation is given below</u>:

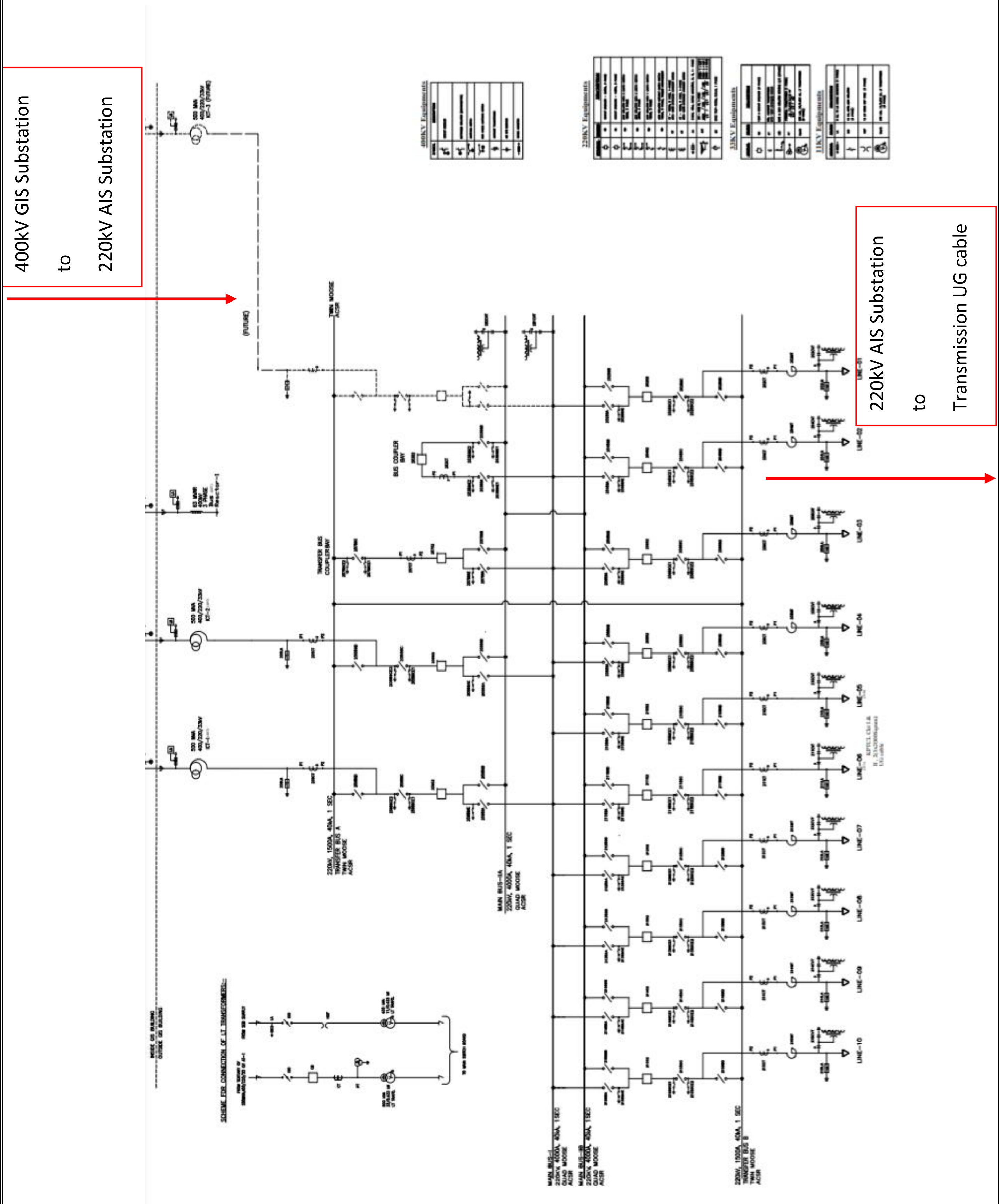


Fig 6.3. SLD of AIS 220kV

➢ <u>TAP HERE TO SEE THE COMPLETE SLD</u> .

1.5. **<u>Equipment of AIS 220kV Substation</u>:**

1.5.1. Why Air Insulated Substation?

- Air Insulated Substation (AIS) or Outdoor Substations have all switchgear equipment, busbars and other switchyard equipment installed outside open to atmosphere.
- Advantages of Outdoor Substation (AIS):
    - This type of substation arrangement is best suited for low voltage rating substations (step down substations) and for those substations where there is ample amount of space available for commissioning the equipment of the substation
    - The construction work required is comparatively less to indoor switch yard and the cost of switchgear installation is also low
    - In future the extension of the substation installation is easier
    - The time required for the erection of air insulated substation is less compared to indoor substation
    - All the equipment in AIS switch yard is within view and therefore the fault location is easier and related repairing work is also easy
    - There is practically no danger of the fault which appears at one point being propagated to another point for the substation installation because the equipment of the adjoining connections can be spaced liberally without any appreciable increase in the cost
- Disadvantages of Air Insulated Substation (AIS):
    - More space is required for outdoor substation when compared to indoor gas insulated substation (GIS)
    - Switch yards are more vulnerable to faults as it is located in outside atmosphere which has some influence from pollution, saline environment and other environmental factors. Deposition of saline particles on insulators can cause insulator failures. They are also vulnerable to direct lightning strikes and other external events such as heavy winds, rains and cyclones.
    - Regular maintenance is required compared to indoor substations as they are exposed to outside environment

1. <u>Bus Scheme</u> – Double Main and Double Transfer Bus scheme.

- This is an alternative of a double bus system.
- The main conception of Double Main and Double Transfer Bus System is, here every feeder line is directly connected through an isolator to a second bus called transfer bus.
- The isolator in between transfer bus and feeder line is generally called bypass isolator.
- The main bus is as usual connected to each feeder through a bay consisting of the circuit breaker and associated isolators at both sides of the breaker. There is one bus coupler bay which couples transfer bus and main bus through a circuit breaker and associated isolators at both sides of the breaker.
- If necessary, the transfer bus can be energized by main bus power by closing the transfer bus coupler isolators and then breaker. Then the power in transfer bus can directly be fed to the feeder line by closing the bypass isolator. If the main circuit breaker associated with the feeder is switched off or isolated from the system, the feeder can still be fed in this way by transferring it to transfer bus.
- In this AIS substation, we have:
    i. Transfer Bus A (220kV, 1500A, 40kA, 1s, Twin Moose, ACSR)
    ii. Main Bus IIA (220kV, 4000A, 40kA, 1s, Quad Moose, ACSR)
    iii. Main Bus I (220kV, 4000A, 40kA, 1s, Quad Moose, ACSR)
    iv. Main Bus IIB (220kV, 4000A, 40kA, 1s, Quad Moose, ACSR)
    v. Transfer Bus B (220kV, 1500A, 40kA, 1s, Twin Moose, ACSR)

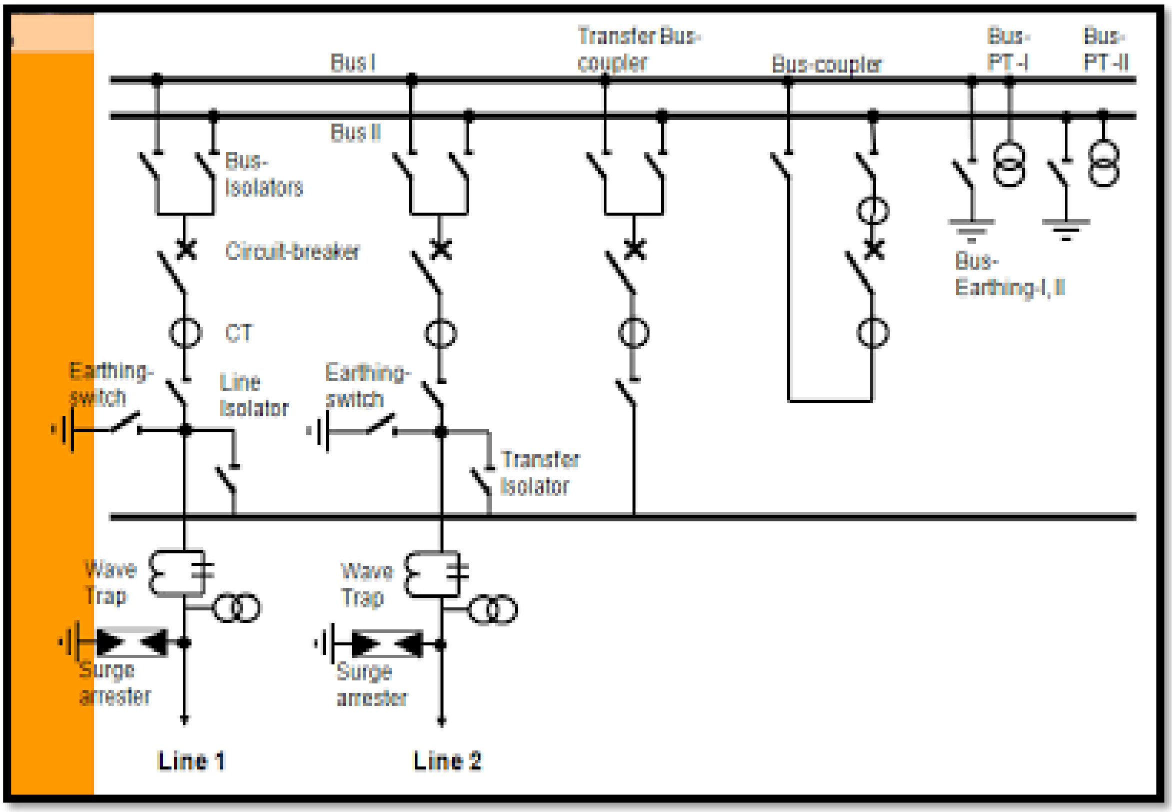


Fig 7. Double Main Bus and Transfer Bus SLD representation

2. ICT – Inter Connected Transformer:

- Manufactured by BHEL.
- 2 nos. (403, 405)
- Rated: HV & IV – 500MVA; LV – 167MVA
- Oil Forced Air Forced cooling type.
- This is a 3-phase 50 Hz Star Connected Auto Transformer (Single winding transformer).
- The HV side is 400kV, IV side is 220kv and LV side is 33kV.
- The 33kV is used as a tertiary winding for auxiliary purposes.
- The neutral point is earthed through Pipe earthing as well to Main earth.
- The ICT has Lightning Arrestors (LA) on both the sides to prevent the transformer from damage due to high lightning surges.
- This ICT has the following parts: RYB primary bushings, RYB secondary bushings, RYB tertiary bushings, Oil filled transformer tank, Conservator, Oil level indicator, Fans, Radiators & Tubes, Buchholz relay, OLTC conservator, 2 breathers (silica gel), OLTC surge relays, Pressure relief valve, On-load tap changer unit, On-line transformer drying unit, Bushing CTs, Marshalling Box (Cooler Control Cabinet), Oil Pockets, Magnetic oil gauge, Earthing, etc.

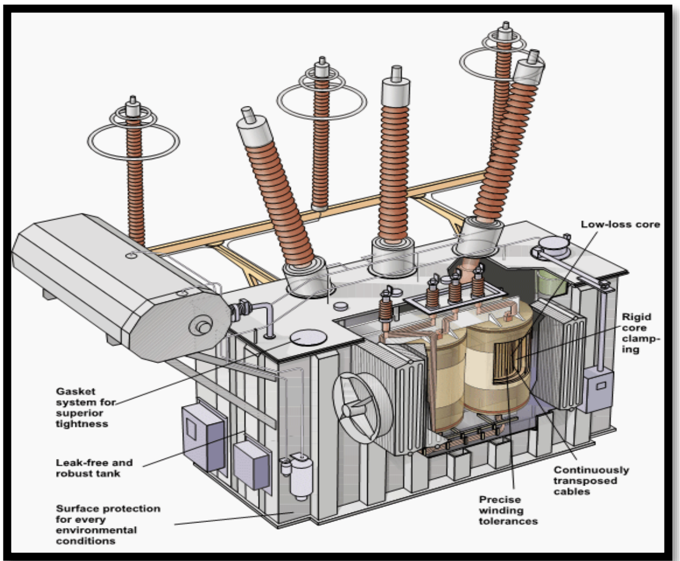


Fig 8.1. Inside an ICT

Fig 8.2. ICT

3. Tertiary Transformer:

- Manufactured by Toshiba.
- 1 no. (800kVA, 33kV/440V)
- It is connected to the LV side of the ICT-2.
- It reduces the unbalancing in the primary due to unbalancing in three phase loads.
- It is required to supply an auxiliary load in different voltage level in addition to its main secondary load. This secondary load is taken from tertiary winding of the ICT-2.
- It assists in limitation of fault current in the event of a short circuit from line to neutral (stabilizes neutral point voltage).

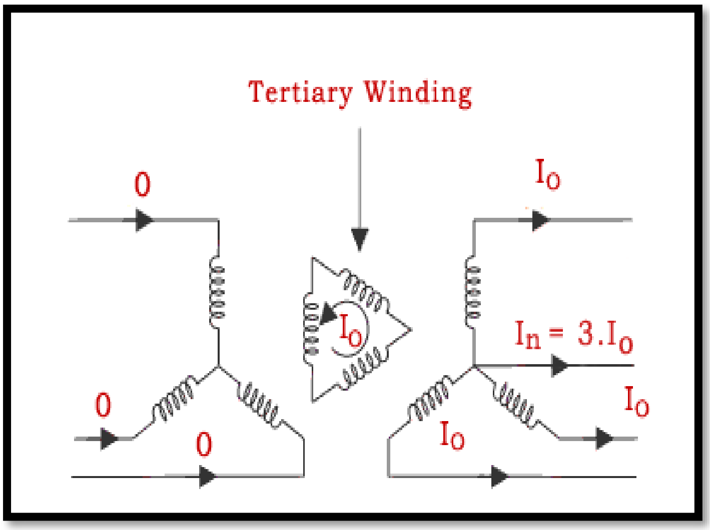
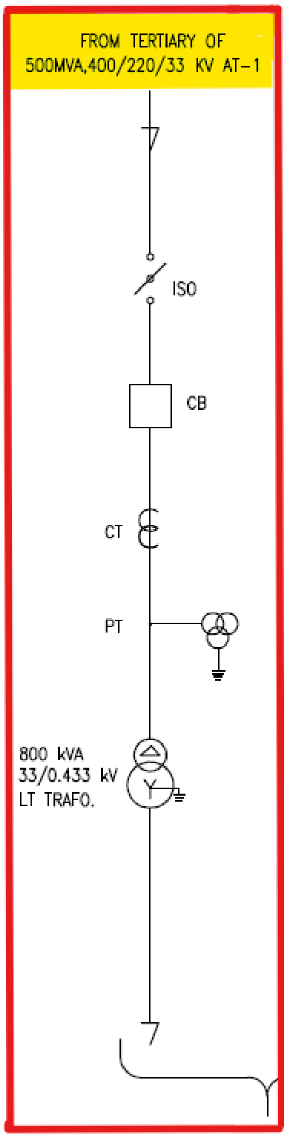


Fig 9.2. Tertiary windings



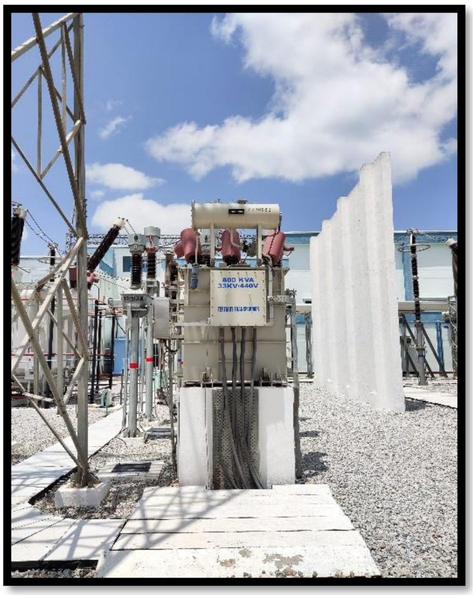


Fig 9.1. Scheme for connection of Tertiary Transformer

Fig 9.3. Tertiary Transformer

4. Circuit Breakers:

- Manufactured by Siemens.
- 14 nos., but 6 no. in running status.
- Rated 245kV, 3150A 50kA.
- SF_6 Type Breaker mechanism; Can be operated during On-Load.
- Fast operation in less than 2 cycles (< 40 milli seconds).
- A circuit breaker in which SF_6 under pressure gas is used to extinguish the arc is called SF_6 circuit breaker.
- SF_6 (sulphur hexafluoride) gas has excellent dielectric, arc quenching, chemical and other physical properties which have proved its superiority over other arc quenching mediums such as oil or air.
- It consists of the following parts: Interrupter unit, moving contact, Moving member, Fixed contact, Arcing horn, Arc, SF6 gas inlet and Gas system.
- The contacts of the breaker are opened in a high-pressure flow sulphur hexafluoride (SF6) gas and an arc is struck between them.
- The gas captures the conducting free electrons in the arc to form relatively immobile negative ions. This loss of conducting electrons in arc quickly builds up enough insulation strength to extinguish the arc.
- Comparing to other circuit breaker types, the SF6 circuit breakers have been found to be very effective for high power and high voltage service.
- SF6 CBs are used in substations for all voltages ranging from 144 to 765 kV or even above. Continuous currents up to 8000 A, and symmetrical interrupting ratings up to 63 kA at 765 kV and 80 kA at 230 kV.

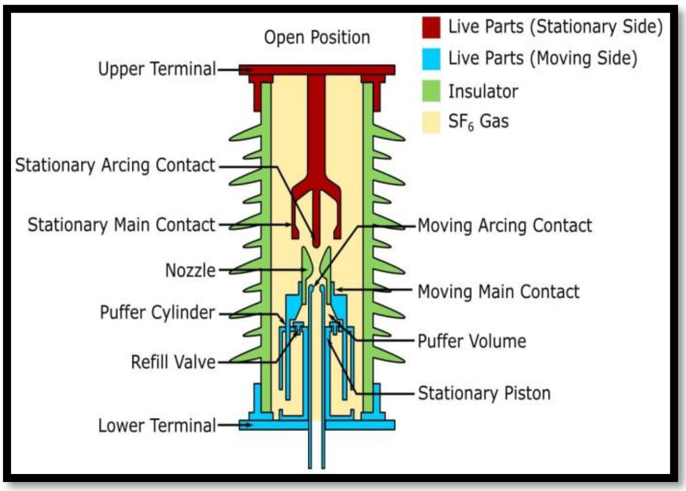


Fig 10.2. Circuit Breaker Parts

Fig 10.1. Circuit Breaker

5. Isolators:

- Manufactured by Siemens.
- 53 nos.
- Rated 245kV, 1600A; Can be operated only during Off-Load.
- Isolators used: Bus isolator, Line isolator, Transfer isolator.
- Isolator type: Double Break Isolator.
- Slow operation in about 10 seconds.
- An isolator is an electrical switching device that disconnects the equipment or electrical lines.
- A mechanical device for providing isolation of power equipment from the network, a disconnector is suitable for switching very small currents or where no significant change in voltage occurs across the terminals.
- The option of earthing sections of power systems can be made available by providing each disconnector pole with one or two earthing switches.
- It can be operated manually or automatically.
- It ensures the total disconnection of the circuit such that no chance of any energizes part in the system.
- DB Iso type has three stacks, first is central or flat male contact which is the movable and the second one is female contact or fixed one which is fitted at both sides of the central one.
- The rotating one comes itself as fixed one to close the circuit and movement of the movable contact out of the fixed one disconnects the system.

Fig 11.1. Isolator

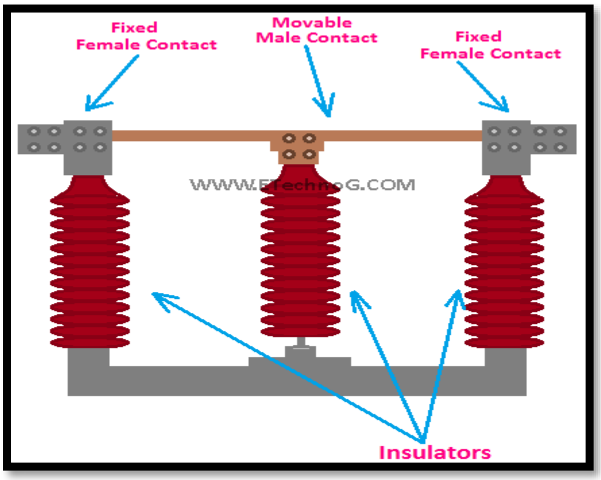


Fig 11.2. Double Break Iso Diagram

6. Earth Switches:

- Manufactured by Siemens.
- 43 Nos.
- Rated 245 kV, 400A.
- Motor type drive mechanism.
- Also called a Single Break Isolator.
- Earthing switch is used to ground the residual charge in power lines after disconnecting the line from source.
- In normal condition, the isolators and circuit breakers are closed but the earthing switch is open.
- During the maintenance period, the circuit breakers and isolators are kept open and earthing switch kept closed.
- Even after isolator operation there may be some residual charges on the bus which may harm the personnel going for maintenance. So before commencing maintenance we have to ground the isolated bus too to avoid any mis happenings.
- Main function of earth switch is to ground the isolated bus/conductor. It is interlinked with isolator, when isolator opens the circuit, earth switch is closed & when isolator closes the circuit, earth switch is opened.
- So, earth switch provides extra safety to the working personnel.

Fig 11.3. Earth Switches

7. <u>Lightning/Surge Arresters</u>:

- Manufactured by LAMCO.
- 36 Nos.
- Rated 216kV, 10kA.
- Used for the protection of equipment at the substations against travelling waves by diverting the abnormal high voltage to the ground without affecting the continuity of supply.
- It is connected between the line and earth, i.e., in parallel with the equipment to be protected at the substation.
- When a travelling wave reaches the arrestor, its sparks over at a certain prefixed voltage.
- The arrestor provides a conducting path to the waves of relatively low impedance between the line and the ground.
- The surge impedance of the line restricts the amplitude of current flowing to ground.
- Lightning arrestor working principle is, once the voltage surge travels throughout the conductor then it reaches the location of the arrestor where it is installed. So, it will break down the insulation of the lightning arrestor for a moment, so voltage surge can be discharged toward the ground. Once the voltage of the system falls under the fixed value, then the insulation will be restored among the ground & conductor. Further, the current flow toward the ground will be stopped.

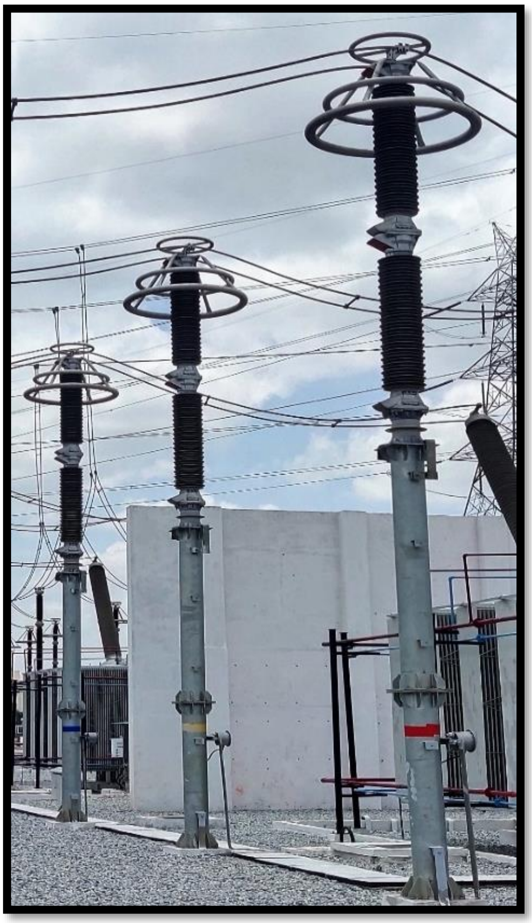


Fig 11.4. Lightning Arrestors

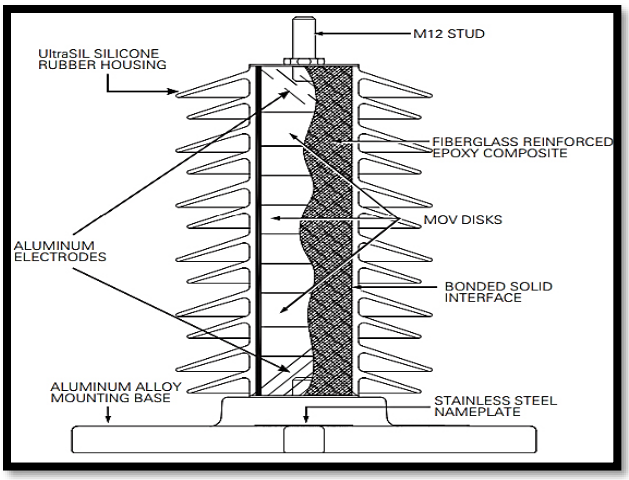


Fig 11.5. Lightning Arrestor Construction

8. Current Transformer:

- Manufactured by Siemens.
- 42 Nos.
- Rated: 245kV, 1600kA.
- Used for the transformation of current from a higher value into a proportionate current to a lower value.
- It transforms the high voltage current into the low voltage current due to which the heavy current flows through the transmission lines are safely monitored by the ammeter.
- It is used with the AC instrument, meters or control apparatus where the current to be measured is of such magnitude that the meter or instrument coil cannot conveniently be made of sufficient current carrying capacity.
- The CT operating principle is based on the law of electromagnetic induction. With a certain number of turns, the voltage from the external network is supplied to the primary power winding and overcomes its total resistance. That results in the appearance of a magnetic flux trapped by the magnetic circuit around the coil. Due to this, there will be minimal loss of electric current during conversion.
- The electromotive force stimulates the magnetic flux at the intersection of the secondary winding switches, located perpendicularly. A current occurs under the control of EMF which is required to resolve coil impedance and output load. At the secondary winding source, simultaneously, a voltage drop is observed.

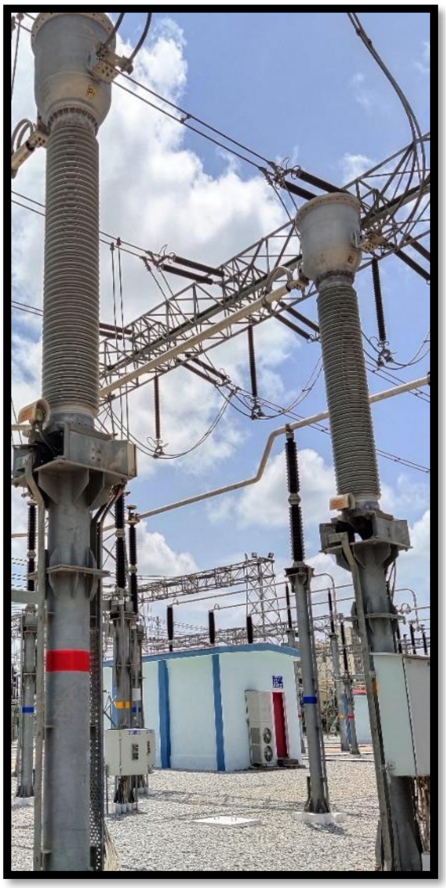


Fig 11.6. Current Transformer

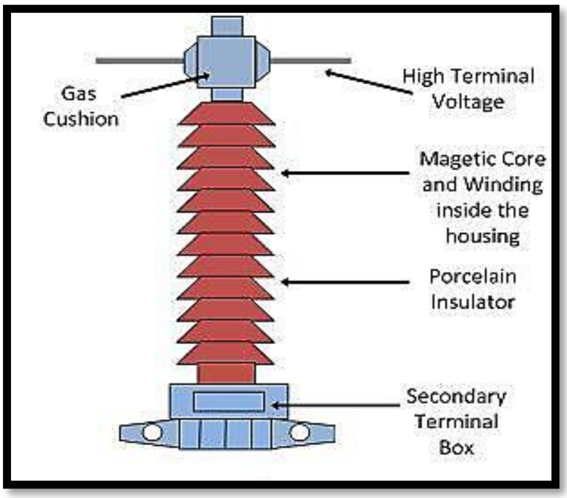
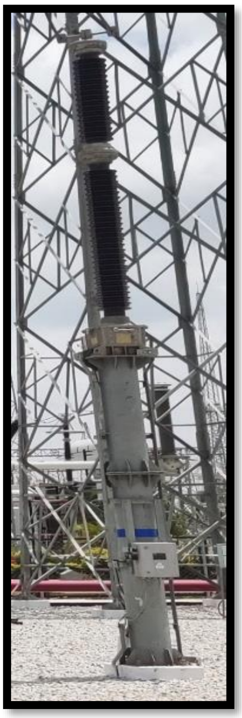


Fig 11.7. Inside a CT

9. Capacitance Voltage Transformer:

- Manufactured by Siemens.
- 36 Nos.
- Rated: 220kV/√3 / 110V/√3
- A switchgear device, used to convert high transmission class voltage into easily measurable values, which are used for metering, protection, and control of high voltage systems.
- It is also used as coupling capacitors for coupling high-frequency power line carrier signals to the transmission line.
- The capacitor voltage transformer is more economical than an electromagnetic voltage transformer when the nominal system voltage increases above 66 kV.



- CVT is working under principle of potential divider. It consists of two capacitors to form a potential divider, line reactor and a step down transformer. Here line reactor is used to compensate the capacitor's phase shift. The value of inductances is adjustable. The inductance compensates the voltage drops occurs in the transformer because of the reduction of the current from the potential divider. But, in actual practice, the compensation is not possible because of the inductance losses.
- The capacitor or potential divider is placed across the line whose voltage is be measured. Let the C1 and C2 be the capacitor placed across the transmission lines. C1 capacitor is nearer to the transmission line and C2 is nearer to the ground. The output of the potential divider acts as an input to the step-down transformer. The capacitor places near to the ground C2 have high capacitances as compared to that placed near the transmission line. C2 > > C1. The high value of capacitances means the impedance of that part of the potential divider becomes low. Thus, low voltages pass to the Potential transformer.

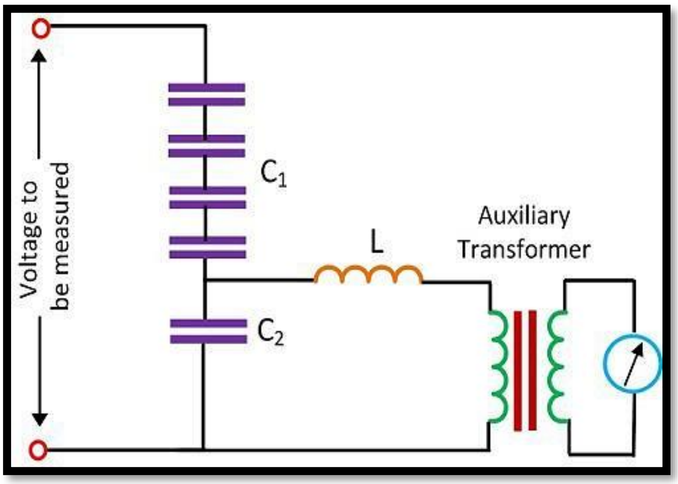


Fig 11.8. CVT parts

10. Wave Trap:

- Manufactured by BPL.
- 16 Nos.
- Rated: 1600A, 0.5 mH
- Wave trap is used to create a high impedance to the carrier wave high-frequency communication entering into unwanted destinations typically substation.
- Carrier wave communication uses up to 150kHz to 800kHz frequency to send all the communication. These high-frequency damages the power system components which are designed to operate 50 or 60 Hz.
- In conjunction with a coupling capacitor, they act as a filter to divert the high frequency carrier signal (in kHz) to/from telecommunication equipment and power frequency signal to/from the substation.
- These high frequency signal should not be coming on the buses as these may damage the equipment.
- A wave trap is a device that allow only a particular frequency to pass through it that it filters the signals coming on to it. So, a wave trap is connected between buses and the transmission line which allow only 50 Hz signal to pass through it.
- This is relevant in power carrier communication (PLCC) system for communication among various substation without dependence on telecom company network. The signals are primarily teleportation signal and in addition, voice and data communication signal.

Fig 11.9. Wave Trap

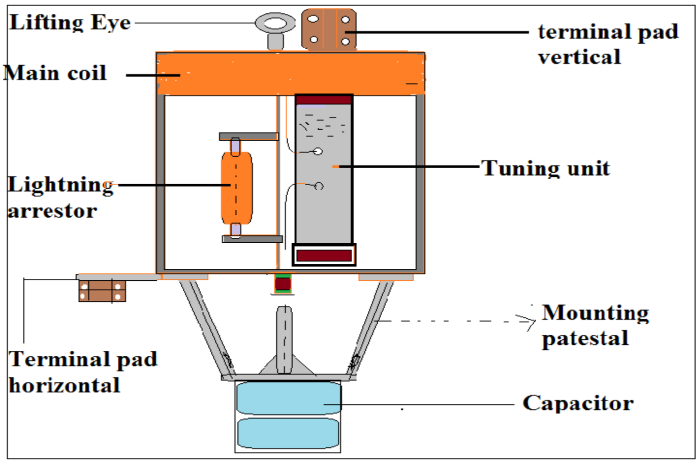


Fig 11.10. Inside a Wave Trap

1.6. Equipment of AIS 400kV Substation:

1.6.1. Transmission Lines:

- 4 incoming feeder lines:
    a. Tumkur-1
    b. Tumkur-2
    c. Nelamangala
    d. Devanahalli
- Minimum Ground clearance of 7.6m.
- Minimum Side Clearance of 77m on either side.

Fig 12. Connection of Transmission Lines to the 400kV AIS substation

1.6.2. Receiving Bay Equipment:

- Each of these 4 lines have a receiving bay consisting of:
    a. Lightning Arrestor
    b. Capacitance Voltage Transformer
    c. Wave Trap
    d. Bus Reactors
    e. Air SF6 bushing

1.6.3. From here, through the Air SF6 bushings, the 400kV AIS is connected to the 400kV GIS.

1.7. <u>Equipment of GIS 400kV Substation:</u>

1.7.1. Why Gas Insulated Substation:

- Gas Insulated Substations (GIS) differ from Air Insulated Substations, such that all the substation equipment such as busbars, circuit breakers, current transformer, potential transformers and other substation equipment are placed inside SF6 modules filled with SF6 gas.
- Generally, Gas Insulated Substations (GIS) are indoor type as it requires 1/10th of the space required for conventional substations.
- Advantages of GIS Substation:
  - It occupies very less space (1/10th) compared to ordinary substations.
  - Most reliable compared to Air Insulated Substations, number of outages due to the fault is less.
  - Maintenance Free.
  - Can be assembled at the shop and modules can be commissioned in the plant easily.
- Disadvantages of GIS Substation:
  - Cost is higher compared to Ordinary Conventional Substations
  - Care should be taken that no dust particles enter compartments which result in flashovers into the
  - When fault occurs internally, diagnosis of the fault and rectifying this takes very long time (outage time is high)
  - SF6 gas pressure must be monitored in each compartment, reduction in the pressure of the SF6 gas in any module results in flashovers and faults.
- Some of the places where GIS Substation are preferred are:
  - Large towns where space available is limited.
  - Industry complexes where uninterrupted power is necessary
  - Mountain regions and valleys
  - Underground substations
  - Off-shore (On sea or lake) substations.
  - HVDC transmission system terminal substations

1.7.2. Why SF_6 Gas?

- Sulphur hexafluoride is an inert, nontoxic, colourless, odourless, tasteless, and non-flammable gas consisting of a sulphur atom surrounded by and tightly bonded to six fluorine atoms.
- It is about five times as dense as air.
- SF6 is used in GIS at pressures from 400 to 600 kPa absolute. The pressure is chosen so that the SF6 will not condense into a liquid at the lowest temperatures the equipment experiences.
- SF6 has two to three times the insulating ability of air at the same pressure. SF6 is about 100 times better than air for interrupting arcs.
- It is the universally used interrupting medium for high voltage circuit breakers, replacing the older mediums of oil and air.
- SF6 decomposes in the high temperature of an electric arc, but the decomposed gas recombines back into SF6 so well that it is not necessary to replenish the SF6 in GIS.
- The SF6 in the equipment must be dry enough to avoid condensation of moisture as a liquid on the surfaces of the solid epoxy support insulators because liquid water on the surface can cause a dielectric breakdown.
- However, if the moisture condenses as ice, the breakdown voltage is not affected.

- So, dew points in the gas in the equipment need to be below–10°C.
- Absorbents inside the GIS enclosure help keep the moisture level in the gas low, even though over time, moisture will evolve from the internal surfaces and out of the solid dielectric materials.
- SF6 is a strong greenhouse gas that could contribute to global warming, but in GIS the SF6 is contained and can be recycled.

Fig 13.0. SF6 Gas Cylinders

1.7.3. Bus Scheme – Double Main Bus scheme.

- This configuration is supplied with two busses. Each circuit is equipped with a single breaker and is connected to both buses using isolators.
- A tie breaker connects both main buses and is normally closed, allowing for more flexibility in operation. A fault on one bus requires isolation of the bus while the circuits are fed from the opposite bus.
- The double bus single breaker scheme is more expensive and requires more installation space than the single bus configuration.
- It is common to find this scheme with an additional transfer bus in EHV transmission substations.
- Four isolators i.e., DS-A, DS-B, DS-C are DS-D are used.
- The Disconnect Switch (DS), often called Isolator, which is connected to Bus-I and Bus-II are called Main Isolator. Thus, DS-1A, DS-1B, DS-2A, DS-2B, DS-3A and DS-3B are main isolators. The DS connected to the load end is called Load Isolator viz. DS-1C and DS-2C. Earth Switch is connected toward load side of Load Isolator. DS-1D and DS-2D are called Transfer Isolators as they directly connect the load to Bus-II.
- In this GIS substation, we have:
    i. Bus Bar Main-1 (400kV, 4000A, 40kA/1 sec, 50Hz)
    ii. Bus Bar Main-2 (400kV, 4000A, 40kA/1 sec, 50Hz)

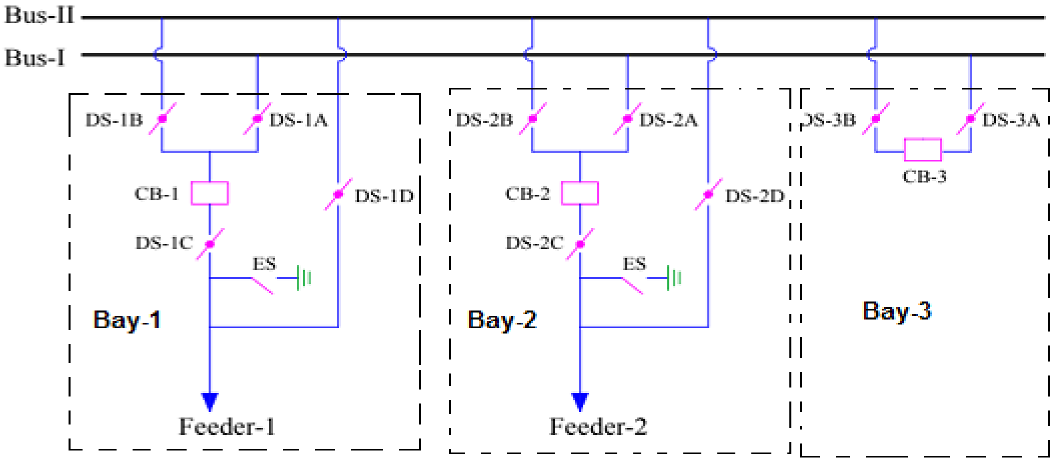


Fig 13.1. Double Main Bus SLD representation

1.7.4. Bus Reactors:

- Manufactured by ALSTOM, BHEL
- 3 nos.
- Rated, 63MVAR, 400kV, 3phase.
- Oil Forced Air Forced cooling type.
- There are 3 Bus Reactors that are connected to the Bus-1 & Bus-2.
- These BRs are placed outside the GIS building, which are in turn connected to the GIS Bus-1 & Bus-2 through the Air SF6 bushings.
- Looks similar to the ICT but has only 3 bushings, whereas the ICT has 6 bushings.
- Bus reactor, an oil-core inductor, is connected between two buses or two sections of the same bus in order to limit the effects of voltage transients on either bus.
- Used to maintain system voltage when the load on the bus changes by releasing reactive power and reduce fault current.
- It consists of the following parts: RYB bushings, Neutral Bushing, Oil filled transformer tank, Conservator, Oil level indicator, Fans, Radiators & Tubes, Buchholz relay, Oil flow control valve, breather (silica gel), OLTC surge relays, Hydrogen moisture (ODGA), Pressure relief valve, On-load tap changer unit, On-line transformer drying unit, Bushing CTs, Turret CT, Marshalling Box (Cooler Control Cabinet), Oil Pockets, Magnetic oil gauge, Earthing boss, etc.

Fig 13.2. Bus Reactor

1.7.5. Incoming Terminals to GIS Building:

- The Incoming terminals from the 400kV AIS are connected to the Air SF6 bushings which are connected to the GIS Busbar.

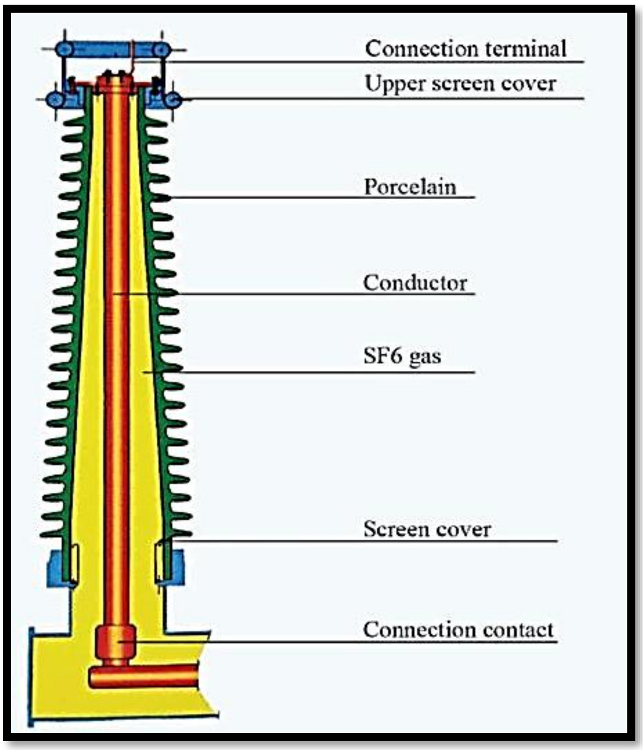


Fig 13.3. Air SF6 Bushings

- For air-to-gas termination, the use of composite insulator for bushings has been gaining importance because they are light-weight, and offer better mechanical and seismic performances.
- A potential transformer, used for metering and protection, forms a part of the GIS & is gas insulated. This equipment is directly mounted & connected to GIS, with a disconnector in series.
- The substations are also equipped with surge arresters to provide additional safety and reliability. The conventional yard surge/lighting arresters are used for gas insulated substation systems, where overhead lines are used to source/deliver the energy.
- The Incoming Terminals to the Bus Bar are:
  - Tumkur-1 line incoming
  - Tumkur-2 line incoming
  - Nelamangala line incoming
  - Devanahalli line incoming
  - Bus Reactors 1, 2, 3 incomings

1.7.6. Bus Bars and Bus Coupler:

- Bus coupler is a breaker used to couple two busbars without any interruption in power supply and without creating hazardous arcs, and also in order to perform maintenance on other circuit breakers associated with that busbar.
- There is one bus coupler bay which couples transfer bus and main bus through a circuit breaker and associated isolators at both sides of the breaker. If necessary, the transfer bus can be energized by main bus power by closing the transfer bus coupler isolators and then breaker. Then the power in transfer bus can directly be fed to the feeder line by closing the bypass isolator. If the main circuit breaker associated with the feeder is switched off or isolated from the system, the feeder can still be fed in this way by transferring it to transfer bus.

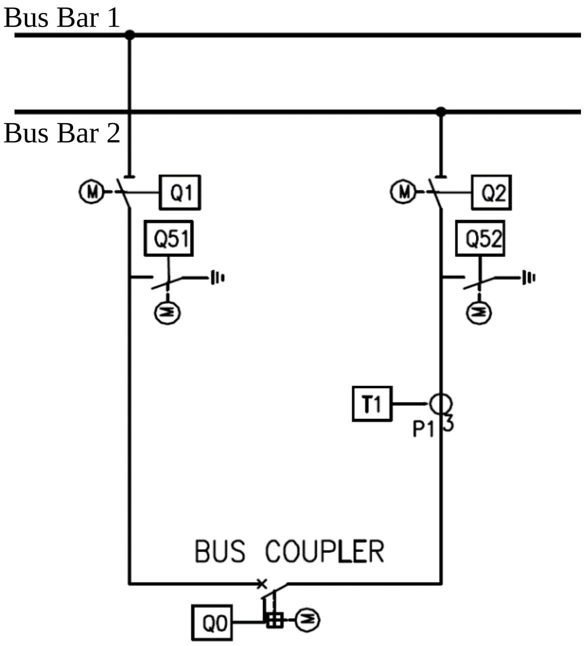


Fig 13.4. Bus coupler Unit and SLD

Fig 13.5. Busbar-1 (Left) & Busbar-2 (Right)

1.7.7. Local Control Cubicle:

- Manufactured by Siemens.
- A local control cabinet (LCC) or Local Control Panel (LCP) is usually provided for each circuit breaker position
- The control and power wires for all the operating mechanisms, auxiliary switches, alarms, heaters, CTs, and VTs are brought from the GIS equipment modules to the LCC using shielded multiconductor control cables.
- The local control cabinet has all necessary control switches, local/off/remote lockable selector switches, close and open switches, measuring instruments, all position indicators for circuit breakers, disconnect switches and grounding switches, alarms, mimic diagram, AC and DC supply terminals, control and auxiliary relays, etc.
- Also, for making operations practicable and convenient as well as trouble-free wiring from GIS, to the substation control room, a LCC is provided.

Fig 13.6. Local Control Cubicle

1.7.8. Disconnectors:

- Disconnectors (or disconnect switches) are placed in series with the circuit breaker to provide additional protection and physical isolation. In a circuit, two disconnectors are generally used, one on the line side and the other on the feeder side. Disconnect switches are designed for the interruption of small currents, induced or capacitively coupled.
- During the closing operation, this gap is bridged by the moving contact. The moving contact is attached to a suitable drive, which imparts the desired linear displacement to the moving contact at a pre-determined design speed.
- A firm contact is established between the two contacts with the help of spring-loaded fingers or the multi-lam contacts. The isolation gap is designed for the voltage class of the isolator and the safe dielectric strength of the gas.
- An insulator is used to drive the moving contact and to isolate the drive from the high voltage components of the disconnector. The shape and size of the insulator are controlled by the electrical and mechanical requirements of the isolator. In three-phase ac systems, the individual phase isolators are ganged together to operate simultaneously.
- The operating speed of the disconnector moving contact ranges from 0.1 to 0.3 m/sec.

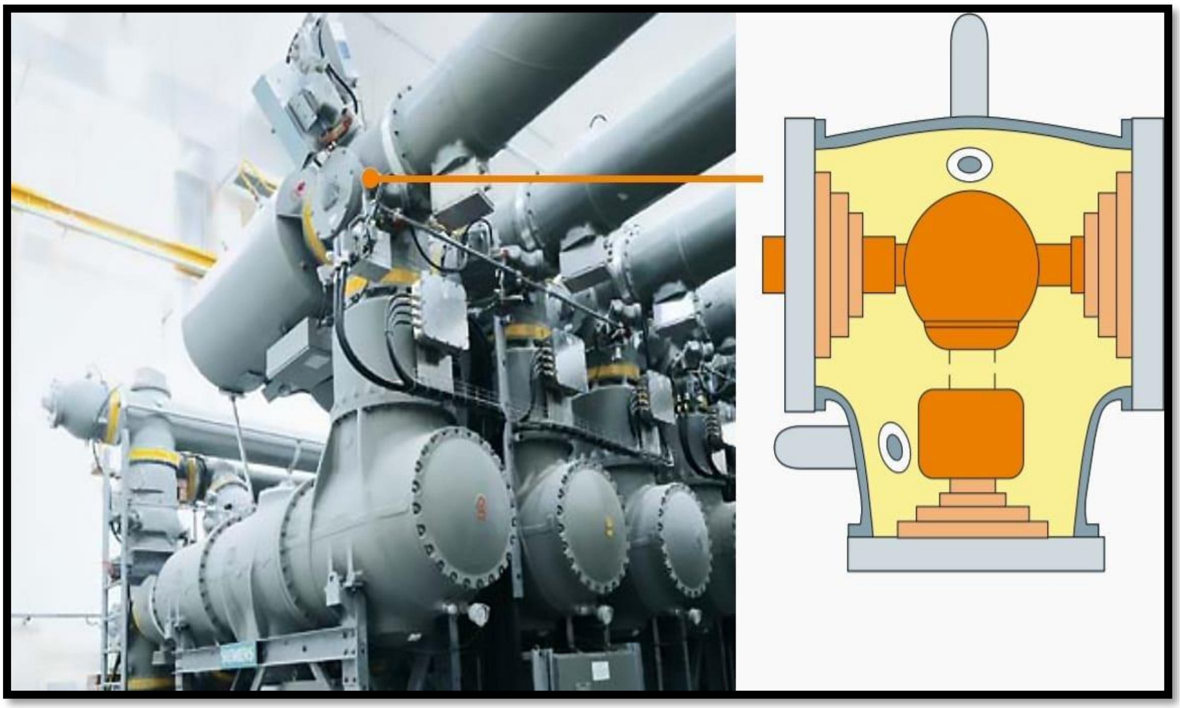


Fig 13.7. Cross section of an Isolated Phase GIS Disconnector

1.7.9. Circuit Breaker:

- The circuit breaker in a gas insulated system is metal-clad and utilises SF_6 gas, both for insulation and fault interruption.
- The SF_6 gas pressure in a circuit breaker is 0.65 MPa.
- The circuit breaker is directly connected to current transformers and the isolators in gas. A barrier is maintained between the circuit breaker and the other connected equipment, operating at lower gas pressure, to maintain a pressure difference.
- Spring, spring-hydraulic and pure hydraulic are the preferred drives for the circuit breakers of gas insulated substations.
- The spring drives are relatively cheaper and can be used only with the state-of-the-art self-blast or hybrid circuit breakers.
- Opening speeds in the range of 6.0-8.0 m/sec and operating energies in the range of 4500-8500 Nm are for operating the GIS circuit breakers.
- The circuit breaker enclosure also serves as the main support element for the individual GIS bay.
- Arc-quenching principle: The interrupter unit used in the circuit breaker for arc-quenching operates according to the dynamic self-compression principle. This principle requires only little operating energy, which keeps the mechanical stresses on the circuit breaker and its housing as well as the foundation loads to a minimum.

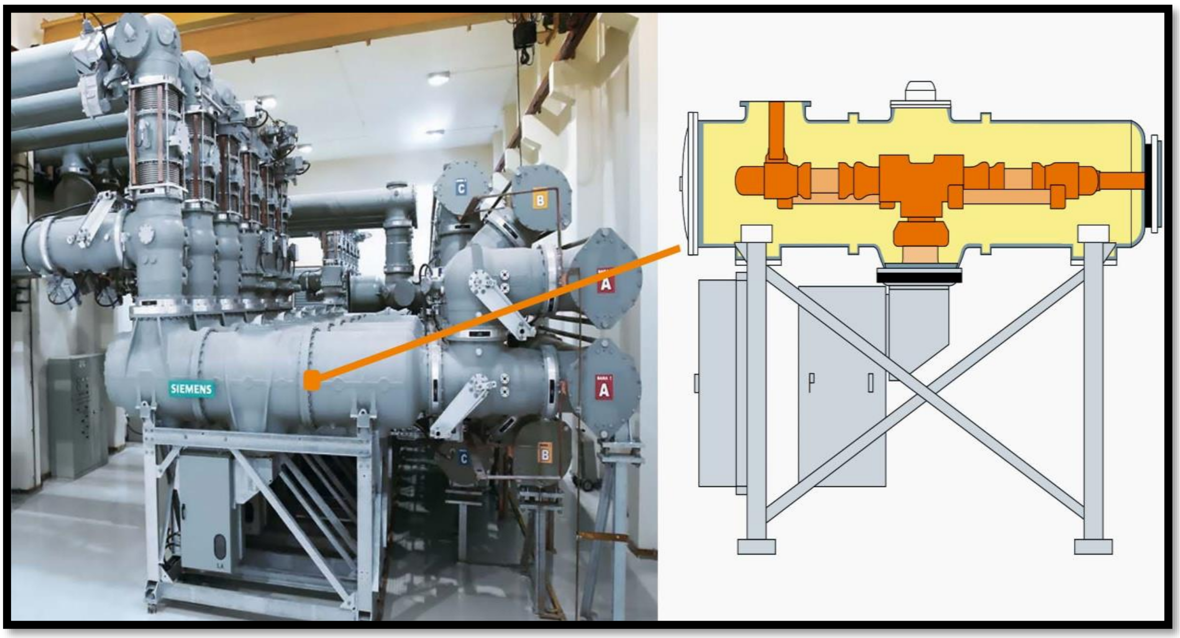


Fig 13.8. Cross Section of a Circuit Breaker

1.7.10. Current Transformers:

- The current transformers in gas insulated systems are essentially in-line current transformers.
- Gas insulated current transformers, with classical coaxial geometry, consist of the following parts:
  - Tubular primary conductor
  - Electrostatic shield
  - Ribbon-wound toroidal core and
  - Gas-tight enclosure
- A ribbon-wound silicon steel core (formed in toroidal shape) is used for the magnetic circuit of the current transformer. A coaxial electrostatic shield, at ground potential, is placed between the high voltage primary and the toroidal magnetic core of the current transformer for ensuring zero potential at the secondary of the current transformer.
- The electrostatic shield also helps in generating a perfect coaxial geometry and uniform electrical field in the gas gap.
- The length of the current transformer module thus changes with the number and types of current transformers specified. The magnetic core and the secondary winding assembly of the current transformer are supported in gas by an enclosure or a grounded support enveloping the core and the winding.

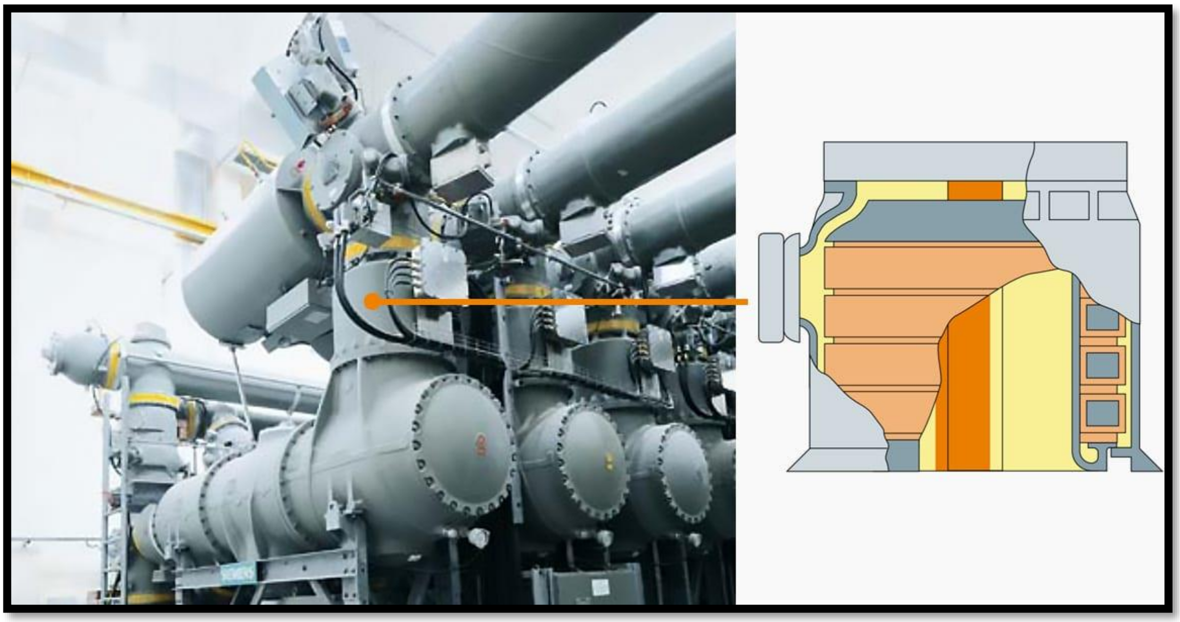


Fig 13.9. Current Transformer in GIS

1.7.11. Earth Switches:

- Fast earth switch and maintenance earth switch are the two types of earth switches used for gas insulated substation systems. The maintenance earth switch is a slow device used to ground the high voltage conductors during maintenance schedules, in order to ensure the safety of the maintenance staff.
- In such a situation, the use of a fast earth switch provides a parallel (low resistance) path to drain the residual static charge quickly, thereby protecting the Potential Transformer from the damages that may be caused.
- The earth switch is the smallest module of a gas insulated substation system. The module is made up of two parts:
    - Fixed contact, which is located at the live bus conductor and which forms a part of the main gas insulated system;
    - Moving contact system mounted on the enclosure of the main module and aligned to the fixed contact.

1.7.12. Outgoing Terminals from GIS Building:

- The outgoing terminals from the Busbars from the GIS building are:
    - ICT – 1 terminals
    - ICT – 2 terminals
- These terminals are taken out of the GIS building and are again connected to the Air SF6 bushings to connect it to the ICT’s, which are placed in AIS.

P.T.O

1.8. Differences Between GIS and AIS:

- Ownership (Total cost of having the equipment):
  - GIS is more likely to cost between ten percent to forty percent more than AIS.
- Maintenance:
  - GIS need less maintenance, inspection every 4 years, and regreasing after at least 20 years.
  - AIS must be inspected every 1-2 years and every compartment should be assessed.
- Operation:
  - GIS are simpler to care and operate as their interior elements are insulated and are provided with integrated test instruments.
  - AIS need more care while operation compared to GIS.
- Setting up:
  - Installation of GIS is easier and quicker because it is smaller, weighs less and takes less space
  - Installation of AIS is difficult and time consuming as the equipment parts are huge and require more time to transport and install.
- Structure:
  - The gas insulated system utilizes a gas, sulphur hexafluoride for insulation.
  - AIS utilizes air insulation within the metal clad system.
- Design Standards:
  - AIS has: IEC (e.g., IEC 62271-1, IEC 62271-200, IEC 62271-100, etc) and media utilized for switching in AIS is air, SF6, vacuum or oil.
  - GIS has: IEC (e.g., IEC 62271-1, IEC 62271-200, IEC 62271-100 …etc) and the media utilized for switching in GIS is SF6 or vacuum
- Size:
  - GIS is smaller.
  - AIS is bigger, as clearances are in the air.

- Sensitivity to Pollution:
    - GIS gas tank offers high ingress protection with IP67 with less sensitivity to humidity, altitude, and environment.
    - In AIS, Humidity and pollution can have impacts on insulation.
- Cable termination requirements:
    - GIS needs particular cable terminations, to keep proper insulation with lowered air clearances in cable connection compartment.
    - AIS doesn’t need any particular requirements. Normal cable termination types could be utilized.
- Type of execution with switching devices
    - GIS has fixed switching devices.
    - AIS have drawn out devices according to design.
- Modularity
    - GIS modularity depends on the design of the range, and cannot be upgraded or extended
    - AIS has modular by nature of the design and provides more flexibility toward upgrades and extension
- Need for gas pressure monitoring
    - GIS might need for both switching devices and busbars integrating SF_6 gas.
    - AIS might need only for switching devices integrating SF_6 gas.

1.9. Other Important Equipment of the Substation:

1.9.1. Earth Mat:

- Designed by Siemens.
- High voltage installations require a grounding system to protect human life against excessive step and touch voltages and to keep transferred potential to a minimum.
- The increase of fault currents to ground affects the importance of grounding systems and the need for low resistance of the ground grid.
- Earth Mats are connected to the Following:
  - The neutral point the system through its independent earth.
  - Equipment framework and other non-current carrying parts of the electrical equipment in the sub-station.
  - All extraneous metallic frameworks are not associated with equipment. Handle of the operating pipe.
  - Fence, if it is within 2 m from earth mat.
- The ground should ideally have 0 Ω resistance, but practically must have < 5.0 Ω, so the substations must:
  - Have Neutrals for generators, transformers, capacitors, and reactors a connection to the earth.
  - Offer a Low-resistance path to the earth for the currents resulting from ground faults, lightning rods, surge arresters, gaps, and other devices.
  - Limit the potential differences that appear between the substation metallic objects or structures, and the ground potential rise (GPR), due to the flow of ground currents; they may pose a danger to equipment and personnel.
  - Improve the operation of the protective relay scheme to clear ground faults
  - Increase the reliability & availability of the electrical system.
  - Allow the grounding of de-energized equipment during maintenance

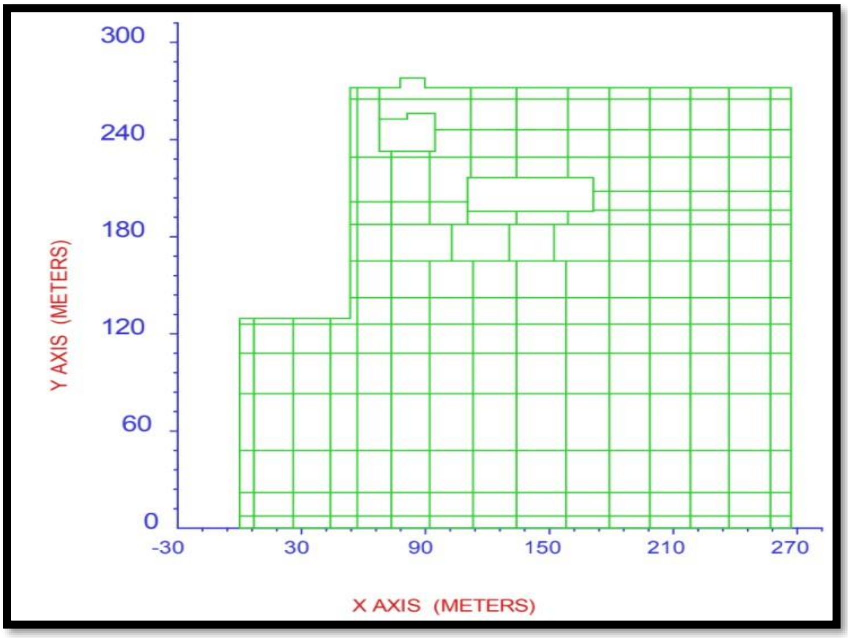


Fig 14.1. Earth Mat of substation

1.9.2. Galvanised steel structures:

- Towers, gantries and equipment supports.
- Hot dip galvanizing is used on these structures to make them weather proof and strong. Galvanizing not only makes these substation structures strong but also corrosion resistant which is very important for any steel or iron structures.
- A transmission tower is a structure set up for the purpose of transmitting and receiving power, radio, telecommunication, electrical, television and other electromagnetic signals.
- Gantry structures are mainly used for guiding the power conductor from last tower near substation to the electrical equipment in a substation. This structure consists of a number of columns and Girder beams, which depend on number of circuits of the line. Gantries are also used for line crossing.

Fig 14.2. Gantry Structures

Fig 14.3. Tower Structures

Fig 14.4. Other Galvanised Steel Structures

1.9.3. Lightning Masts:

- Lightning masts are an effective way to protect a wide variety of fixed and portable installations from lightning damage.
- Lightning masts are usually not mounted to the object that they protect. They are designed to project well above the protected area, and are place near, but not touching the protected objects.
- Lightning currents are conducted away into a ground system, rather than into the object.
- Lightning masts provide a much greater degree of protection to controllers, electrical devices, and other critical.
- They may also be combined with traditional structural systems to provide "spot" protection to HVAC systems, communications gear, and other structure mounted assets.

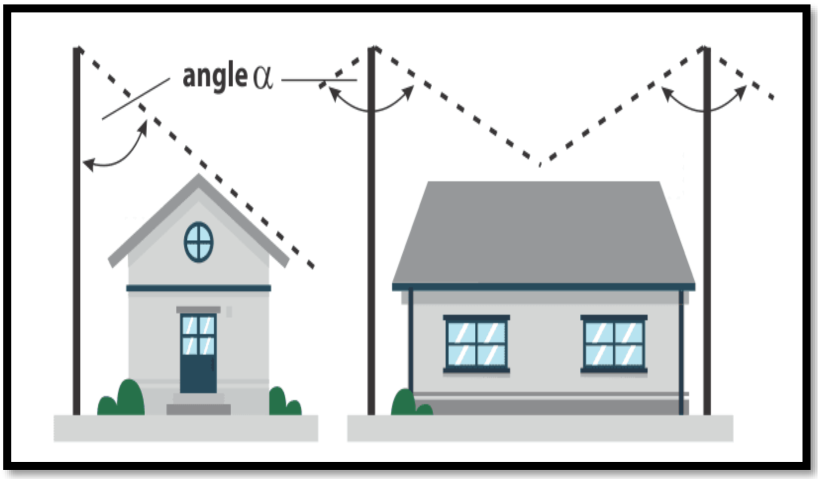


Fig 14.5. Example of area protected under a Lightning Mast

Fig 14.6. Lightning Mast

1.9.4. Cables:

- Electrical cable classification can be done in 2 ways:
- On the basis of insulating material used in their manufacture:
  - Belted cables — up to 22 kV.
  - Screened cables — from 22 kV to 66 kV.
  - Pressure cables — beyond 66 kV.
- On the basis of voltage for which they are manufactured:
  - Low-tension cables — up to 1000 V.
  - High-tension cables — the operating voltage of high-tension cables is up to 11000 V.
  - Super-tension cables — the operating voltage of super tension cable is from 22 kV to 33 kV.
  - Extra high-tension cables — from 33 kV to 66 kV.
  - Extra super voltage cables — beyond 132 kV.

Fig 14.7. Underground cable to KPTCL

Fig 14.8. 2000sq.mm XLPE cable

Cable cross sectional view

1.9.5. Conductors:

- Transmission line conductors are responsible for carrying power from generating station to receiving stations.
- **AAC** – All Aluminium Conductor
  - It contains one or more aluminium alloy strands.
  - Preferred for short spans (usually in urban areas).
  - Effective against corrosion problems in coastal areas.
  - Highest conductivity to weight but has a poor strength to weight ratio.
  - Technical details: Aluminium grade 1350, hard drawn H19 temper.
  - Practical usage: Local distribution levels.
- **AAAC** – All Aluminium Alloy Conductors
  - Alloy conductor that is made of aluminium, silicon, and magnesium.
  - Its ampacity is equivalent to AAC and possesses excellent tension characteristics.
  - They are stronger, lighter and more conductive than ACSR.
  - Technical details: 6201-T81 alloy
  - Practical Usage: 36 kV Transmission and above
- **AACSR** – All Aluminium Alloy Conductors Steel Reinforced
  - They are less common than AAACs.
- **ACAR** – Aluminium Conductors Alloy Reinforced
  - Combination of aluminium strands that are helically wrapped around aluminium alloy wires. The combination increases the mechanical strength of conductors.
  - Technical details: Aluminium 6201-T81 alloy and 1350 aluminium.
- **ACSR** – Aluminium Conductors Steel Reinforced
  - The central steel core is surrounded by a number of aluminium strands. Steel strands, coated with zinc are used for increasing the strength of conductor.
  - Practical usage: Longer spans and High voltage transmission lines.

- **Standard Sizes of Conductor for Lines of Various Voltages:**
- The following sizes have now been standardised by CEA for transmission lines of different voltages:
    - For 132kV lines: 'Panther' ACSR having 7-strands of steel of dia 3.00 mm & 30-Strands of Aluminium of dia 3.00 mm.
    - For 220kV lines: 'Zebra' ACSR having 7-strand of steel of dia 3.18 mm and 54-Strands of Aluminium of dia 3.18 mm.
    - For 400kV lines: Twin 'Moose' ACSR having 7-Strands of steel of dia 3.53 mm and 54-Strands of Aluminium of dia 3.53 mm.

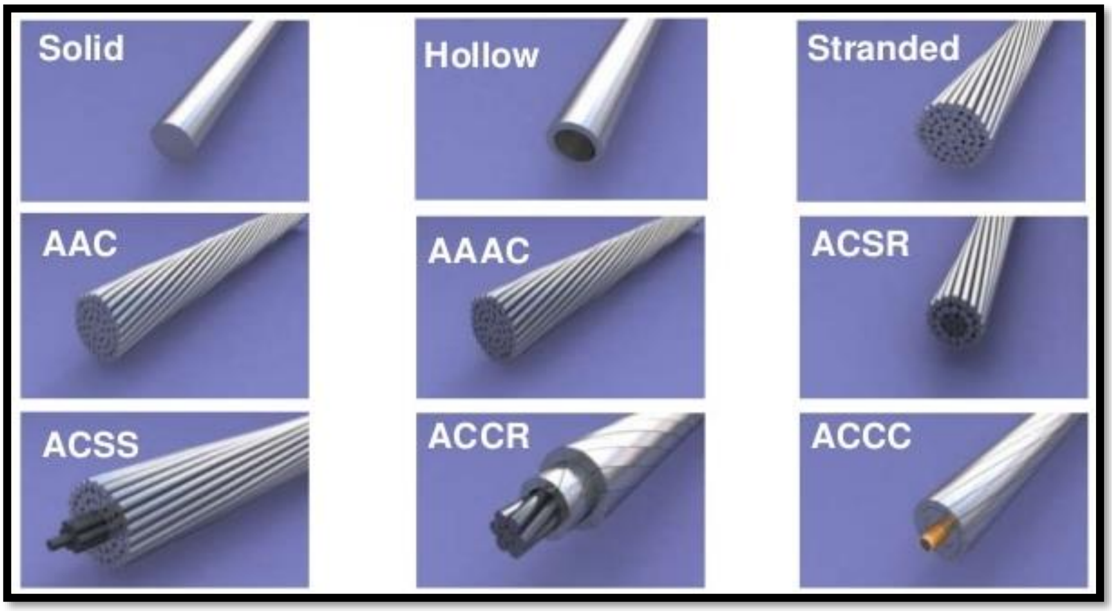


Fig 14.9. Types of Conductors

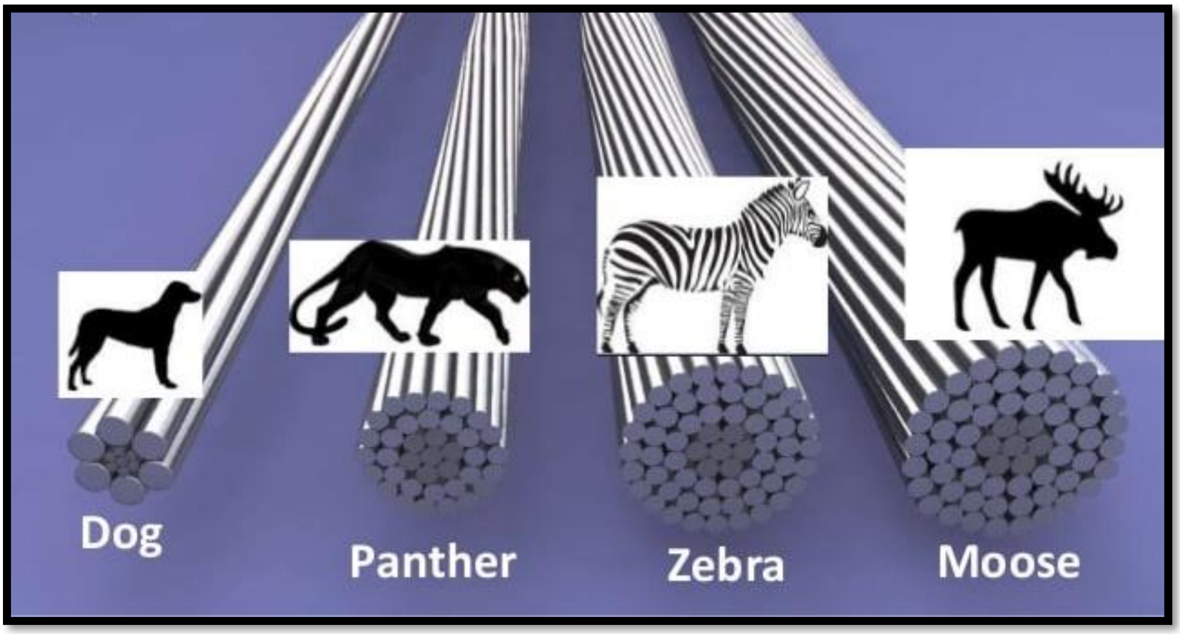


Fig 14.10. Types of ACSR Conductors

1.9.6. Control and Relay Panels: (CRP)

- Manufactured by Siemens, BHEL, ALSTOM
- CRPs are designed to control several feeders, through medium voltage indoor and outdoor switchgear in a primary distribution substation.
- It is typically deployed when associated switchgear does not have space to accommodate the protection, control and monitoring needs of a substation.
- The CRPs can also be extended to incorporate substation HMI, time synchronization, metering and networking functionality in the substation and thereby avoid requirement of a separate control room.
- By deploying CRP solutions also enhances the safety to the substation engineer, as exposure to a live switchgear is mitigated.
- Used for protection, control. measurement & monitoring of electrical equipment such as transformers, generators, busbars, cable or line feeders, bus couplers, capacitor banks, voltage protection etc.
- The CRP systems can include pre-wired, pre-engineered and tested cluster of Relion protection relays, SSC600, SMU615 merging units, RIO600 IO units, auxiliary relays, meters, AC500/AC800 PLCs, PML630, ZEE600, GPS time server, Ethernet switches, semaphore indicators, annunciators, indication lamps, push buttons test blocks/plugs, terminal blocks, DIN rails etc.

Fig 14.11. Control & Relay Panels

1.10. PLCC

1.10.1. Power Line carrier communication:

- Uses Electrical conductors of EHV/HV Transmission Lines for propagation of carrier signals, thereby does not require any additional wire.
- Oldest form of Communication.
- Modulated communication signals on EHV Phase conductors through coupling equipment.

1.10.2. Elements of PLCC system:

- Line Traps
- COUPLING DEVICES
- COUPLING CAPACITOR / CAPACITOR VOLTAGE TRANSFORMER (CVT)
- COAXIAL CABLE
- CARRIER TERMINAL
- POTECTION COUPLER
- PABX

1.10.3. Communication System Requirement:

- Information to be transmitted in the form of signals.
- Device to convert the data/voice signals in to carrier signals suitable for transmission over the selected medium.
- Medium for transmission (telephone lines, copper conductors, electrical phase conductors, earth wires, fibre optic cables etc).
- Device to retrieve information from the carrier signals.

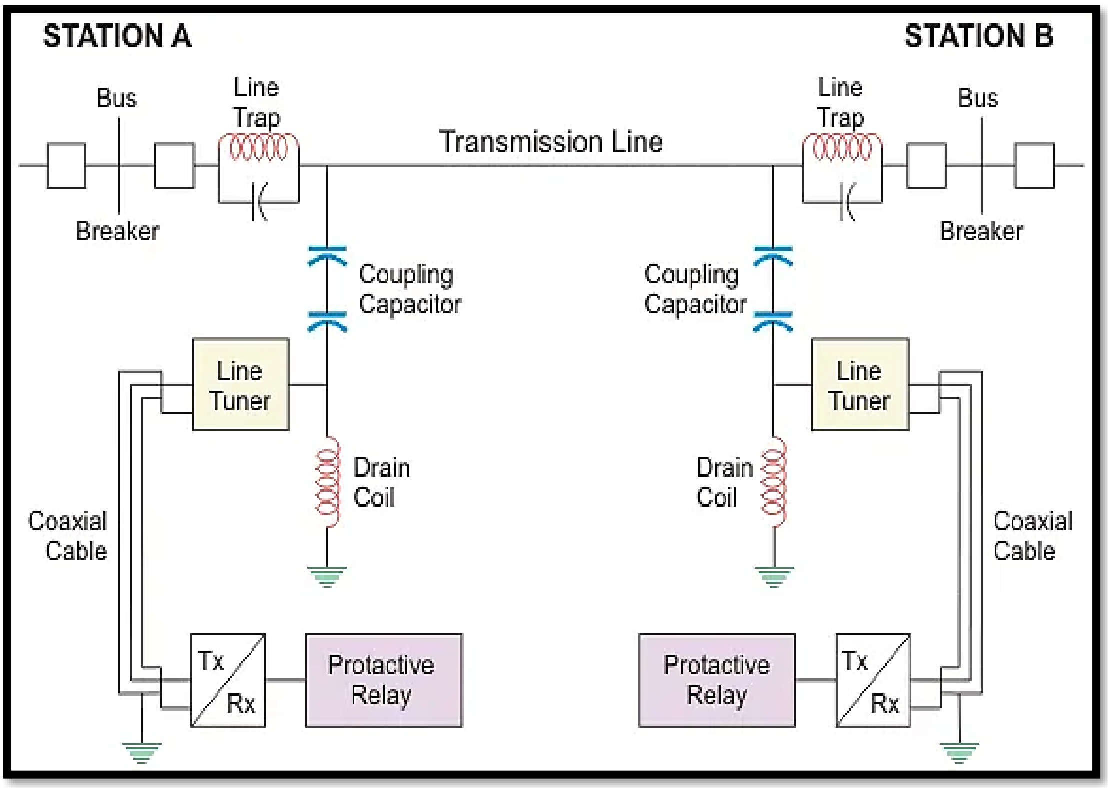


Fig 15.1. Diagram of PLCC network

1.10.4. **Frequency Ranges in PLCC System:**

- Audio:
  - 4khz audio band is used for transmission voice & data.
    - Voice: 0.3 –2.0 kHz
    - Data: 2.1 – 3.4 kHz
- Carrier frequency band
  - 30 kHz - 500 kHz high frequency range is used for carrier signals
- Speech & data signals will be transmitted in the 4khz bandwidth in each direction over the carrier.
- E.g.:
  - With 100/104 kHz carrier frequency, the transmit signals will be 100-104 kHz and receive signals will be 104-108 kHz range.

1.10.5. **Functions of Various Elements in PLCC System:**

- Line traps:
  - A device directly connected in series with a phase conductor.
  - Offers negligible impedance to power frequency currents.
  - Offers high impedance to carrier signals for defined blocking band of line traps.
  - Line trap ratings
    - 0.5mH /2000A, 0.5 mH /1000 A
    - 1.0 mH/2000A, 1.0 mH/1000 A.
    - mH will be selected based on the requirement of blocking band.
    - Current rating will be selected based on Power Flow in EHV line.

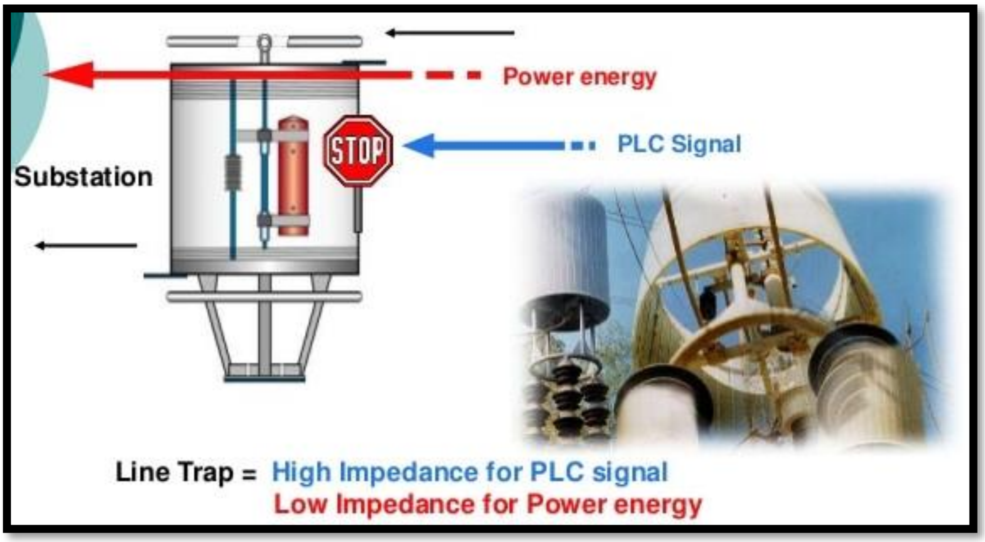


Fig 15.2 Line Trap Signal Blocking

- Coupling capacitor / CVT:
    - This provides coupling between EHV/HV line and communication equipment.
    - Typical ratings are 3300/4400/6600/8800 pf.
- Coupling device:
    - Consists of surge arrestor, drain coil, matching transformer, earth switch.
    - Provides impedance matching
    - This is band pass filter. The band depends on capacitance of CVT.
- Coaxial cable:
    - Connects the carrier terminal in the control room to coupling device in the switchyard.
- Carrier terminal:
    - Basic communication equipment to connect 0.3-3.4 kHz voice & data signals
- Protection coupler:
    - This is used to interface the protective relay with the carrier terminal.

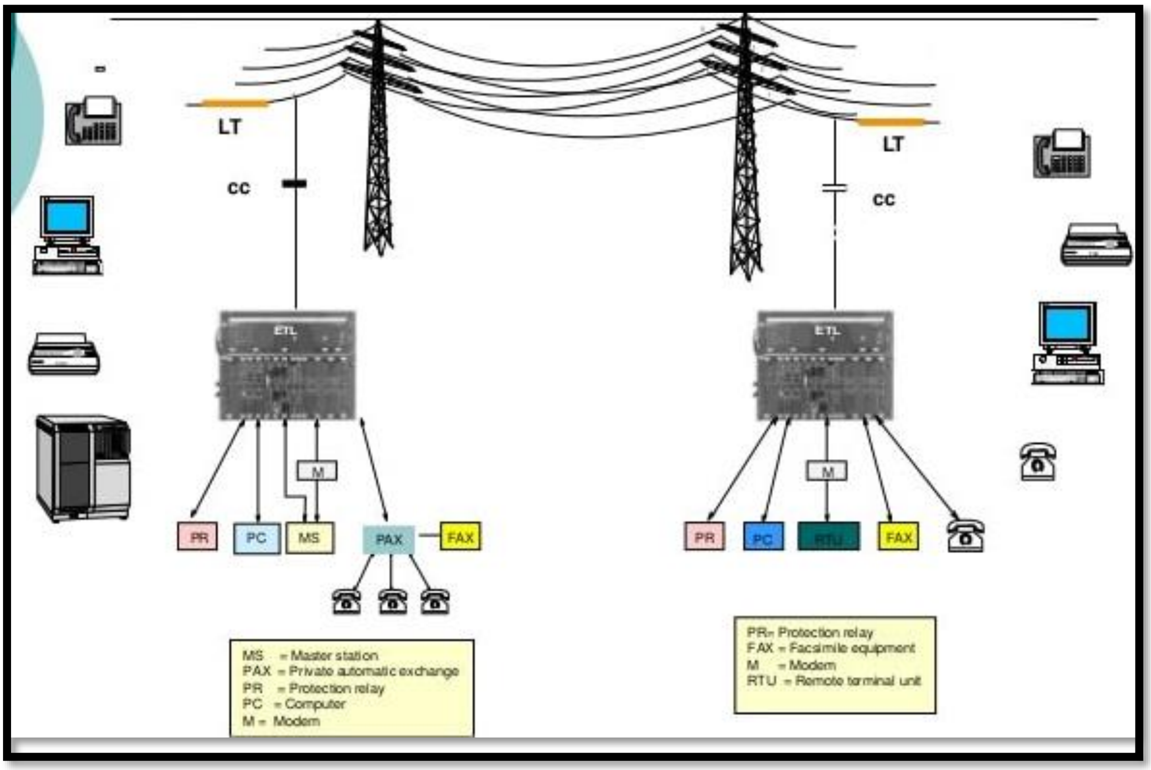


Fig 15.3 Typical PLCC Installation

1.10.6. **Coupling in PLCC System**:

- Phase to ground coupling
- Phase to phase coupling
- Inter-line coupling

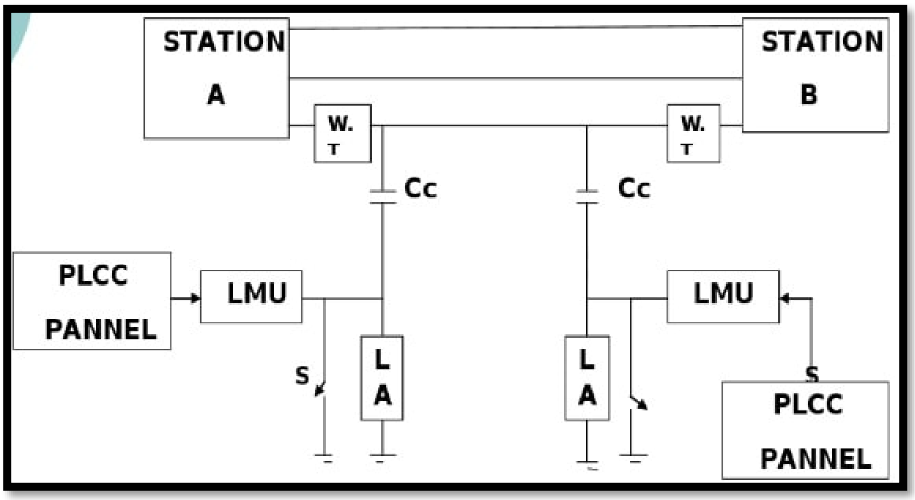


Fig 15.4 Phase to Ground Coupling

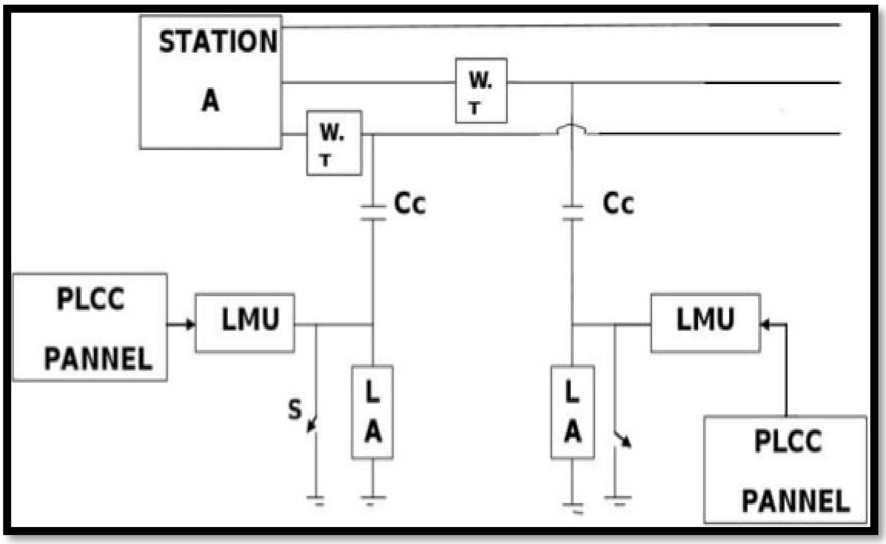


Fig 15.5 Phase to Phase Coupling

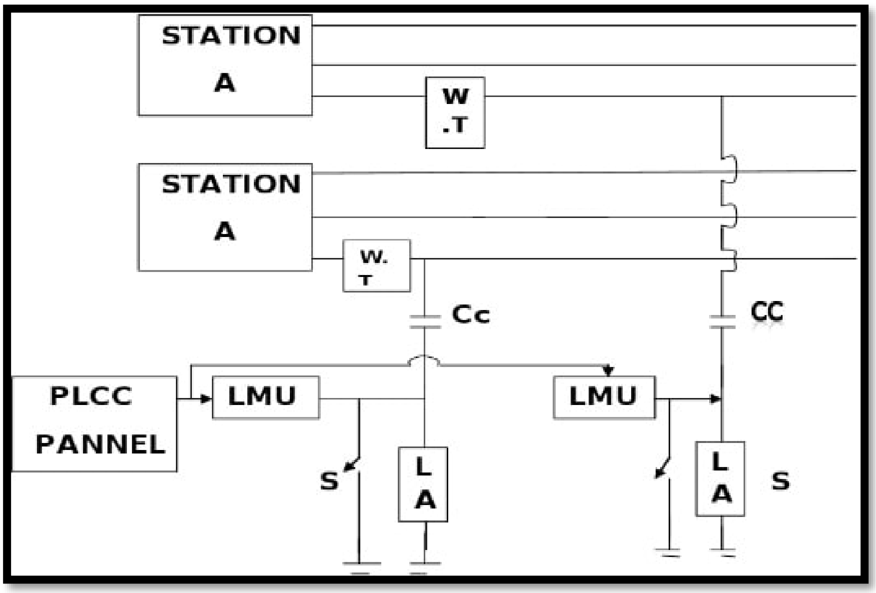


Fig 15.6 Inter-Line coupling

1.11. Fire Fighting System:

- Has 2 pipes:
    - Red pipe: Water flow pipe
    - Blue pipe: Detection of fire (pressure) pipe
- When fire occurs, the Bulbs in the Blue pipes break, causing the pressure to decrease.
- If the pressure goes below 8 kg/cm^3, the sensor will detect it and water will be pushed out through the Red Pipes on to the Equipment.
- Emulsifier systems are placed near the ICTs and Reactors.
- Hydrant points are also placed in multiple locations in the substation as well as the Control Room building to manually extinguish fire.
- A deluge fire protection system has unpressurized dry piping and open sprinkler heads. The system is directly connected to a water supply and when the system is activated, a deluge valve will release the water to all the open sprinkler heads. The valve is opened when activated by a heat or smoke detection system.
- Jockey Pump:
    - A small pump connected to a fire sprinkler system to maintain pressure in the sprinkler pipes.
    - This is to ensure that if a fire-sprinkler is activated, there will be a pressure drop, jockey pump would be activated to overcome this pressure drop.
    - For minute leakages.
    - 2 Nos.
- Electrical Motor Pump:
    - 1 No.
- Diesel Engine Pump:
    - 1 No.
- Air Vessel Tank:
    - The air vessel tank is used as an accumulator to store compressed air, to separate condensate through cooling and to compensate for pressure fluctuations in a compressed air distribution system. In water supply systems, air vessels are used as hydrophores and also as safety components to avoid surge pressure.

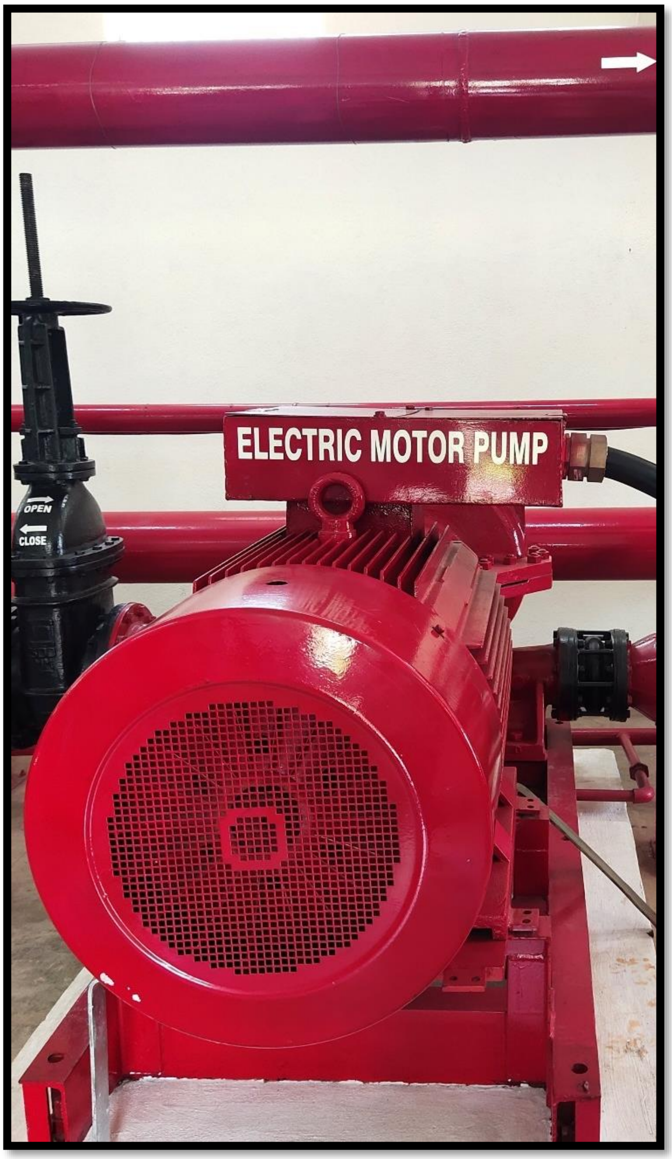
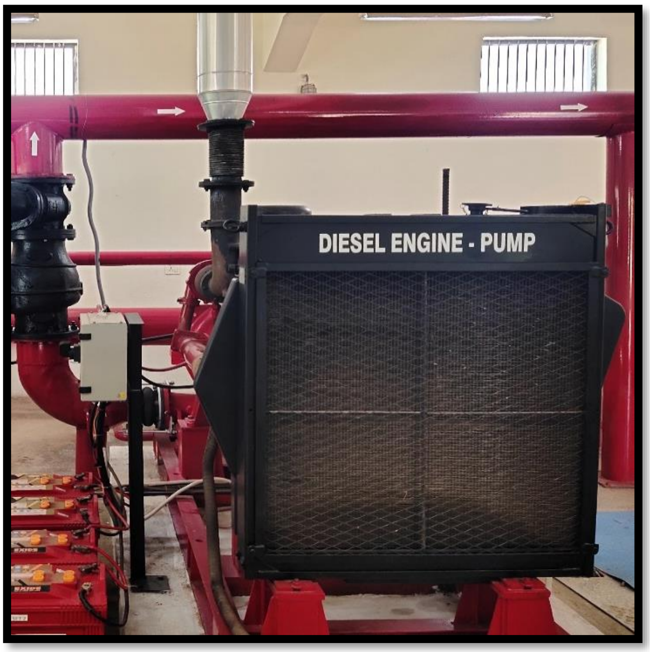


Fig 16.1 Air Vessel Tank

Fig 16.2 Electric Motor Pump

Fig 16.3 Jockey Pumps

Fig 16.4 Diesel Engine Pump



1.12. Battery Room:

- A battery room is a room in a facility used to house batteries for backup or uninterruptible power systems.
- Battery rooms are found in telecommunication central offices, and to provide standby power to computing equipment in datacentres.
- Batteries provide direct current (DC) electricity, which may be used directly by some types of equipment, or which may be converted to alternating current (AC) by uninterruptible power supply (UPS) equipment.
- Batteries Used: Valve regulated Lead Acid Batteries
- Manufactured By: Exide
- Each Rated 2.25V
- Bay 1 & 2:
    - For 220V each.
    - 107 cells connected in series, for each bay.
- Bay 3 & 4:
    - For 48V each.
    - 23 cells connected in series, for each bay.

Fig 17.1. 48V Bay-1

Fig 17.2. 220V Bay-3 & Bay-4

1.13. MCC Room:

- Power distribution systems used in large commercial and industrial applications can be complex. Power may be distributed through switchgears, switchboards, transformers, and panel boards.
- Power distributed throughout a commercial or industrial application is used for a variety of applications such as heating, cooling, lighting, and motor-driven machinery.
- In many commercial and industrial applications, quite a few electric motors are required, and it is often desirable to control some or all of the motors from a central location. The apparatus designed for this function is the motor control centre (MCC).
- There are 2 Distribution Boards:
- AC DB:
    - It has 3 sources: BESCOM (11kV), ICT (33kV), DG set (415V)
    - Used for lighting, office purpose, relay supply.
- DC DB:
    - Source: Batteries, Charger – 48V/220V
    - Goes to relays supply and also to charge the batteries.
    - Charging Battery is done using Float cum Boost Charge method.
    - Under normal condition FCBC works as a float Charger.
    - When the mains fail, battery takes over supply to the load.
    - On resumption of power, FCBC switches to the “Boost Mode,” Boost charges the discharged battery and returns to the “Float Mode”, after the battery is restored to full charge.
    - All along, it supplies uninterrupted DC power to the load.

Fig 18.1. 48V Battery Charger

Fig 18.2. 220V Battery Charger

Fig 18.3. 415V AC distribution board

2. RTAMC, SR-II

Regional Transmission Asset Management Centre, POWERGRID, Yelahanka

(Also functions as Back Up NTAMC)

- Total no. of Substations in Southern Region-II: 33(HVAC); 3(HVDC)
- Total Transformation capacity: 30,787 MVA (72 ICTs)
- Total No of Transmission Lines: 147 (14764.405 ckt kms)
- Total No. of Bus Reactors: 33 (3153 MVAR)

2.1. What is NTAMC:

- National Transmission Asset Management Centre.
- Centralised control of entire transmission system from single point with fully automated remote controlled sub-stations.

2.2. Why NTAMC & RTAMC?

- Technical Challenges:
  - With merging grids (North - East – West – North East & South), inter-regional power flow needs more coordination.
  - Long delays in layered system of manual operation.
  - Delays in gathering & evaluating intelligence in stand-alone substations.
- Economic Challenges:
  - Huge Manpower Cost in Manned system.
  - No requirement of manning S/S operation in today's digital, automated control system.
  - Low technology cost in Automation & Remote Control.
  - Gearing up for competitive bidding regime of transmission assets.
  - Stringent demands by the regulator etc.
- To overcome these challenges, Technological development in substation automation, coupled with falling prices of communication system and IT provides the opportunity for VIRTUAL MANNING of the substation thereby optimizing the requirement of skilled manpower and managing the asset with the reduced & quality manpower.

2.3. NTAMC & RTAMC:

- State of the art computerized control Centres NTAMC & RTAMC are being set up by PowerGrid with associated telecom system to:
  - Enable Remote operation of substation
  - Enable Remote monitoring of assets through intelligent online device.

2.4. Objectives:

- Reduction of operation cost through remote operation.
- Reduction of Maintenance cost by Maintenance Service Hubs.
- Better condition monitoring through ONLINE devices.
- Competitive edge for PowerGrid:
  - Reduction of downtime of network by centralized operation
  - Better evaluation of situation through ONLINE intelligence
  - Better coordination with RLDC.
- Safe & wise grid:
  - NTAMC / RTAMC shall be manned round the clock by qualified and skilled staff. Ploughed back through unmanning the S/S Operation.

2.5. Highlights:

- ONE (01) - NTAMC
- NINE (09) - RTAMC (One for each region)
- A Back Up NTAMC & Disaster Recovery Centre at Bangalore.

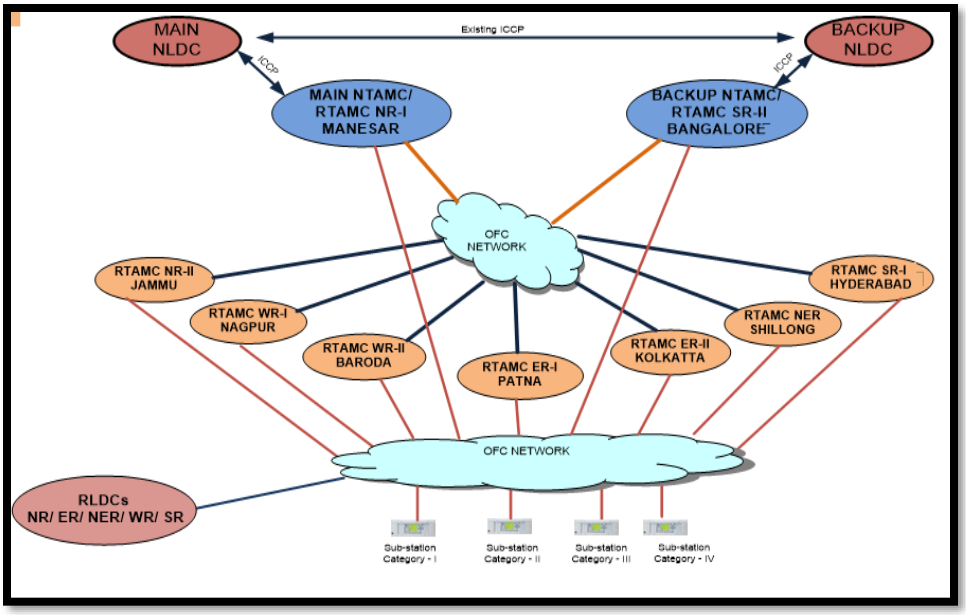


Fig 19.1 NTAMC & RTAMC Communication Architecture

2.6. RTAMC Functions:

- Computer System and Hardware:
  - Consists of SCADA system, Power system Applications, Historical data storage, Interface with ERP system, GPS based image processing application, Web servers etc.
  - Also, Operator Training Simulator is also to be provided at Backup NTAMC.

- Remote Supervision and Control:
  - For Remote Supervision and Control of the Substation from NTMC/RTMC, following functions are envisaged:
    - Measurement
    - Alarm
    - Control
    - Condition monitoring of the equipment
    - Transmission Asset Management
    - Video Surveillance System
- Communication Architecture:
  - Substations and the various control centres will be connected by redundant broad band communication network of POWERTEL and ULDC communication links.
  - At stations where this connectivity is not possible, leased lines will be hired from telecom operators up to the nearest connection point.
- Connectivity Status:
  - All the Locations shall have dual connectivity, to improve reliability of Communication system.
  - Connectivity through POWERTEL network:
  - Connectivity through ULDC FO network.
  - Connectivity through service provider network.
- Video Surveillance System:
  - Real-time video surveillance of substations will be deployed by providing Camera Management Servers at various Control Centres to provide visibility of the substation and its equipment through strategically placed cameras.

- The visual images complemented with data from SCADA system and will strengthen the operating decision.
- The augmentation of the existing CCTV system will be carried out at respective substations for effective utilization of the same for video surveillance system at Control Centres.

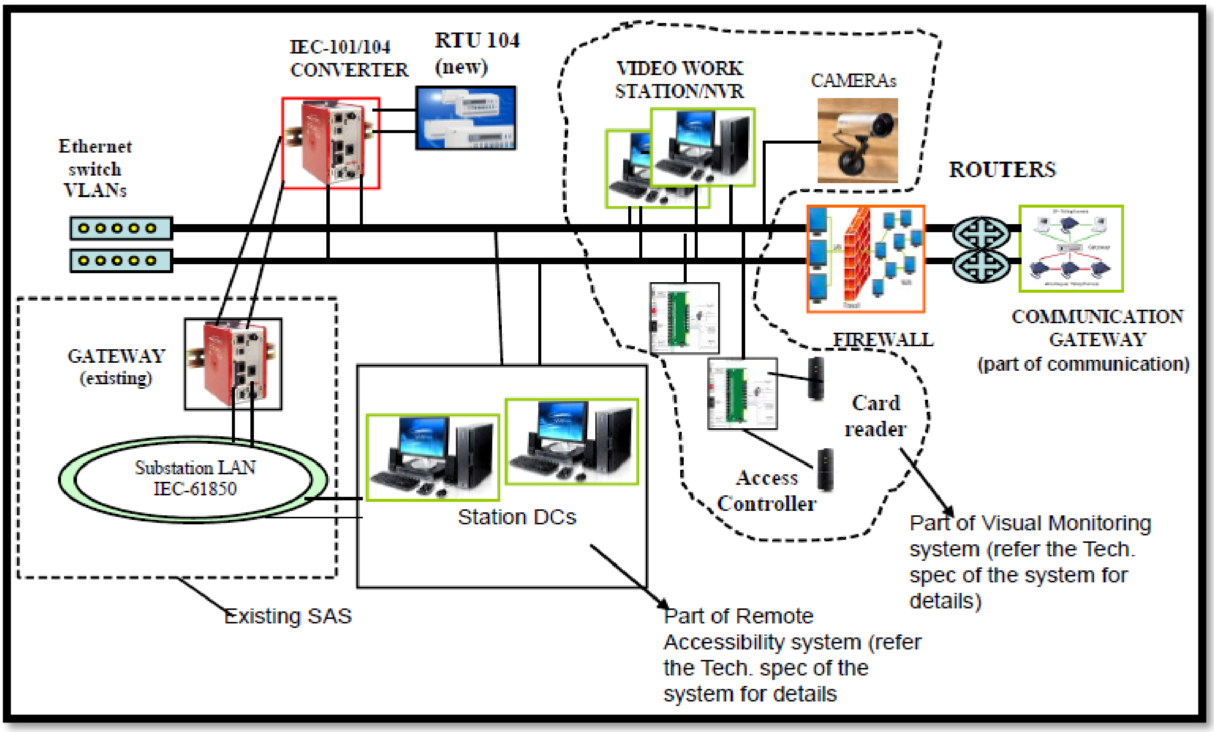


Fig 19.2 Monitoring and Control Map

2.7. RTAMC Functions under CPCC

- "Central Project Co-Ordination Centre".
- Centralized Operation, Monitoring, Asset Management and Optimization of POWERGRID Transmission system.
- Ensuring Safe & Secure Operation of POWERGRID Transmission Assets.
- Coordination with RLDC centrally for system operation & Faster restoration of the transmission elements thus enhancing the availability and reliability of the network and improving response time to the network contingency.
- To enhance the asset life by ensuring operation of equipment within rated parameters.
- Coordinated action in case of Grid Parameters being outside the operational limit.
- Coordination with,
  - Station & RLDC for Shutdown related maintenance activities.
  - Station for non-shutdown related maintenance activities.
- Co-ordination with SLDS (Karnataka, Kerala, Pondy & Tamilnadu).

- Co-ordination with around 90 state utility stations.
- Issue & Cancellation of PTW/SFT to Maintenance/MRT Engineer for S/S Maintenance.
- Coordination with RLDC & SLDC for Shutdown approval and obtaining the opening / charging code for availing the S/D.
- SRLDC to be contacted in the event of any abnormal Grid Condition and in case of any action is warranted to control abnormality, same need to be complied as per specific instruction from SRLDC with proper code.
- Coordination with Telecom department to ensure availability of the MPLS network for data reporting.
- Planned Shut downs of Elements approved in OCC
- Requests for approved shut down in prescribed format has to be collected from concerned stations before three days of the date of approved shut downs and to be uploaded on SRLDC Outage Portal.
- In case of approved shut down for construction activities, the shut down requests have to be collected through PMS from CAOs before three days of the date of approved shut downs and to be uploaded on SRLDC Outage Portal.
- If any deviation of approved dates due to unavoidable reasons, the same will be informed to SRLDC through mail in coordination with stations before 3 days.
- Planned Shut downs of Bay equipment:
- Requests for approved shut down in prescribed format has to be collected from concerned stations before two days of the date of proposed shut downs.
- Bay shut down of the same feeder at both stations should not be permitted for maintenance at the same time.
- If any station is availing Bus shut down, no other bus shut down will be permitted to the connected stations.
- During Bus shut down, no other shut downs such as relay tastings, bay equipment are allowed. This should ensure by the maintenance engineer/Substation In-charge.
- Whiling availing any Bay equipment, it should ensure there is no interruption of power flow in the connected feeders.

2.8. Remote Monitoring Parameters – Analog:

PARAMETER	TRANSMISSION LINE	TRANSFORMER	REACTOR
Active Power	YES	YES	NO
Reactive Power	YES	YES	YES
Current (all 3 phases)	YES	YES	YES
Voltage (all 3 phases)	YES	YES	YES
Frequency	YES	YES	NO
Winding Temperature	NO	YES	YES
Oil Temperature	NO	YES	YES
Tap Position	NO	YES	NO
Online DGA Parameters	NO	YES	YES

2.9. Remote Monitoring Parameters – Digital:

EQUIPMENT	PARAMETER
Circuit Breaker	Position, Interlock Status, SF6 Gas Alarms, Operating System Alarms, AC & DC Supply status, Trip Coils Status
Isolator & Earth Switch	Position, Interlock Status
Main-1 & 2 Relays of Transmission Line	Relay Healthiness, Zone Operation indications, Over Voltage / SOTF / PSB / Stub Protection indications, Carrier send/receive status, Fuse failure, Auto Reclose status, Master Trip indication
Main Relays of Transformer	Relay Healthiness, Differential / REF / Backup Over Current & Earth Fault / Over Flux Protection indications, Fuse Failure status
Main relays of Reactor	Relay Healthiness, Differential / REF / Backup Impedance Protection indications, Fuse Failure status, Master Trip indication
Transformer & Reactor Mechanical Protection	Buchholz / Pressure Relief Valve / Surge Relay / High Winding Temp / High Oil temp /Fire Protection Alarms/ Cooling System status
Main Relays of Busbar	Zone Operation / LBB indication / Relay Healthiness / CT Supervision Alarm
Bay Controller unit	Healthiness, Local/Remote Status, Time Synchronization status

2.10. Local Monitoring Parameters – Analog:

- Circuit Breakers SF6 Gas Pressure
- Circuit breakers Operating Mechanism (Oil/Air) Pressure
- Transformer Conservator Oil Level
- OLTC Conservator oil Level
- Reactor Conservator oil level
- Lightening Arrestor Leakage Current & Surge Counter reading
- Current Transformer Oil level or SF6 Gas level
- Capacitor Voltage Transformer oil or SF6 Gas level
- Bushing Oil level
- CB operation counter readings
- PLCC operation counter readings

2.11. Substation Auxiliary (Fire Fighting System):

EQUIPMENT	PARAMETER
LT SYSTEM	Analog Parameters: - Active/Reactive Power, All three Phases Current & Voltage, Frequency & Power factor Digital Parameters: - CB Position, Spring Charging status, Transformer Buchholz/WTI/OC&EF Protection/Negative Phase sequence/UV indications
DIESEL GENERATOR SET	Analog Parameters: - Active/Reactive Power, All three Phases Current & Voltage, Frequency, Power factor & Battery Voltage Digital Parameters: - CB Position, Spring Charging status, DG Set Alternator fault/Oil Pressure low/Cooling water temp. High/ Engine High Speed / OC&EF Protection/ Negative Phase sequence/UV indications
FIRE FIGHTING PROTECTION SYSTEM	Analog Parameters: - Header Pressure & Battery Voltage Digital Parameters: - Diesel Engine Oil Pressure low/Cooling water temp. High/ Engine High Speed indications, Running & L/R status of Diesel Engine/Electric Pump/Jockey pump/Air Compressor
DC SYSTEM	Analog Parameters: - Charger Current, Battery Bank Voltage & Current Digital Parameters: - Charger Mode, Trouble Alarm, DC Under Voltage & E/F

CONCLUSION

Internships are very important and beneficial because they help in developing our professional aptitude, strengthen personal character, and provide a greater door to opportunity.

We tend to learn the practical approach towards the work, and can co-relate between the theory and practical aspects of the studied topics.

Doing my internship in 400kV/220kV (GIS/AIS) POWERGRID substation has taught me the installation, operation, control, and maintenance of various equipment used in a substation. The different types of tests, methods & procedures, handling, protections for all the equipment, which I observed and studied has enhanced my knowledge and my practical approach ability towards my engineering subjects.

Also learning about PLCC, RTAMC, Fire Fighting Protection, etc., has made it very evident to understand the practical difficulties and the need of all the new technologies in electrical substation.

It was an overall experience with all the officials, engineers and other staff at POWERGRID, Yelahanka; who showed me how to work in a work place and to understand how a company functions.

Thank you all.

REFERENCES

a. https://www.powergridindia.com
b. https://electrical-engineering-portal.com
c. Transmission & Distribution by Uday A. Bakshi (Technical Publications)
d. https://www.elprocus.com
e. https://circuitglobe.com
f. Fundamentals of Modern Electrical Substations by Boris Schwartzberg, Ph.D., P.E., P.M.P.
g. https://new.abb.com/
h. Switchgears book by BHEL – Bharat Heavy Electricals Limited
i. Gas-insulated switchgear up to 550 kV, 63 kA, 5000 A, Type 8DQ1 – SIEMENS